An Anatomy of U.S. Firms Seeking Trademark Registration

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Abstract

This paper reports on the construction of a new dataset that combines data on trademark applications and registrations from the U.S. Patent and Trademark Office with data on firms from the U.S. Census Bureau. The resulting dataset allows tracking of various activity related to trademark use and protection over the life-cycle of firms, such as the first application for a trademark registration, the first use of a trademark, and the renewal, assignment, and cancellation of trademark registrations. Facts about firm-level trademark activity are documented, including the incidence and timing of trademark registration filings over the firm life-cycle and the connection between firm characteristics and trademark applications. We also explore the relation of trademark application filing to firm employment and revenue growth, and to firm innovative activity as measured by R&D and patents.

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1 Introduction

Attracting consumers and retaining them as loyal customers is critical for a firm's survival and growth. Among the many ways of building and protecting a loyal customer base, trademarks are unique. By protecting a firm's intangible assets, trademarks can reduce consumer search and switching costs, lower the expense of introducing and marketing new products, and generate brand awareness and loyalty. While trademarks generally facilitate establishing and enhancing goodwill, they may not always be directly linked to a particular attribute of the firm. Goodwill can be generated through investment in consistent quality, exceptional customer service, a distinctive portfolio of products, or a unique service or innovation that makes a firm stand out. The fact that not all firms file to register trademarks, despite the relatively low cost of doing so, suggests that certain firms stand to benefit more from trademark registration than others.[1]

Theory contends that firms use trademarks to appropriate the returns from investments in goodwill. Firms are therefore more likely to select into trademark registration when the returns to reputation, product quality, and scale and scope expansion are high.[2] Likewise, when innovative activity complements goodwill, by enhancing product or service quality and inducing customer loyalty, firms have more motivation to protect accumulated and anticipated goodwill with a trademark registration. The benefits derived from using a trademark can, in turn, affect firm performance and productivity. When firms successfully leverage trademarks to differentiate goods or services and insulate themselves from copying and competition by registering trademarks, they can achieve faster growth. At the same time, the price elasticity of demand for firms with trademarks can be lower, leading to higher markups. Maintaining higher markups, in turn, may intensify firms' advertising and marketing activities or foster further investment in quality enhancement and process or product innovation. Consequently, protection of trademarks through registration can have long-term consequences for a firm's competitive position in the market as well as the industry concentration.

While theory suggests trademarks can play a critical role in firm dynamics and innovation, empirical research regarding which firms in the United States use trademarks and the benefits they thereby derive is relatively sparse, especially when compared to other intellectual property such as patenting. The small but growing body of empirical literature on trademarks relies

[1] One rationale for why not all firms file to register trademarks is provided by Landes and Posner (1988) pp.271-272: If trademarks signal consistent quality, quality may be costly to maintain, and can be verified by consumers after purchase. Hence, only the firms that can afford to provide such consistent quality will tend to seek trademark registration.

[2] See, e.g., Landes and Posner (1987,1988) and Economides (1988) for theoretical arguments on the connection between trademarks and consumer behavior. For recent models of firm dynamics under costly and gradual customer acquisition, see Dinlersoz and Yorukoglu (2012) and Gourio and Rudanko (2014).

primarily on application and registration data from other countries, particularly the United Kingdom, Australia, and France. Such firm-level studies generally find trademark filing and/or registration to be correlated with product differentiation, marketing, and innovation, though results vary by industry. The prior literature also finds a positive relationship between trademark registration and firm market value, productivity, and survival, indicating that the private value of trademarks to firm is positive, though there is yet no clear conclusion regarding their social value. Still, most firm-level research to date relies on datasets of large, publicly-traded firms, which casts doubt on the applicability of results to the population of firms that seek trademark registration.

One major obstacle to empirical analysis of trademark use by firms in the United States is the lack of comprehensive data on firm-level trademark activity. Recently, however, there has been substantial progress on this front. The United States Patent and Trademark Office (USPTO) has made available trademark data that covers nearly 7 million trademark applications for the period 1870-2015. The USPTO Trademark Case Files Dataset (TCFD) is a remarkable synthesis of various trademark activity by firms.[3] It contains detailed information on applications for trademark registration, as well as the commercial use, renewal, assignment, and cancellation of registered trademarks. It identifies the dates an application is filed with the USPTO and proceeds to registration, and in what product categories or classes of goods and services are covered by a registration. However, the TCFD has little to contribute regarding the characteristics of the firms that seek trademark registration and when exactly in their life-cycle they do so. It is, therefore, not possible to uncover how trademark filing is related to firm characteristics and dynamics with the TCFD alone.

This paper reports on the construction of a new dataset that combines the TCFD with firm-level micro data at the U.S. Census Bureau. The dataset fills a void in the literature by linking trademark activity with firm characteristics, performance, and dynamics in the United States. It provides information on the incidence and timing of trademark filing and registration over the life-cycle of a firm and, thus, opens several research possibilities. The trademark-firm linked data can be used to explore not only what kind of firms seek to register trademarks, but also when they do so, and how trademark filing is related to firm dynamics, such as entry, survival, employment and revenue growth, and R&D and patenting intensity. This paper provides a first look at the connection between trademark filing and broader measures of firm outcomes based on the constructed data. An objective of this initial analysis is to explore some of the selection and treatment effects associated with seeking federal trademark registration in terms of firm growth and innovation.

[3] See Graham, Hancock, Marco and Myers (2016) for details on the construction and features of the USPTO Trademark Case Files Dataset.

Key events early in the life-cycle of firms may signal the emergence of high growth firms and generate skewness in firm outcomes. There is a growing interest in identifying precursors of successful businesses. Recent research indicates that having a patent or a trademark application is highly correlated with the ultimate success of an early entrepreneurship activity, as measured by rare events such as an IPO or a high-value acquisition.[4] Analysis of the constructed data, likewise, indicates that trademark filing is correlated with employment and revenue growth. There appears to be strong selection into trademark filing for trademark registration based on firm size and age, though size is a more critical correlate. Firms that do not apply for trademark registration in their initial years are unlikely to do so unless they experience employment growth. Difference-in-difference analysis suggests sizable treatment effects, with firms seeking trademark registration having substantially higher employment and greater revenue in the period following first filing relative to control firms.

Among the least studied aspects of trademarks is their ability to capture firm innovation. Trademarks can be used to capture the value of firm innovative outputs that are not covered by patents, such as innovations in retail, services, customer relations, and knowledge-intensive products. Little is known, however, about this function of trademarks. More evidence is needed on firms use of trademarks to appropriate returns from innovation and the relationship between trademark activity and more traditional measures of innovative activity. Firm-level analysis of the constructed data indicates that firms with R&D and patent activity are also very likely to apply to register trademarks. Further, the relatively frequent co-presence of trademark applications and R&D activity in firms without patents suggests that, for at least some firms, trademarks may capture innovative outputs of R&D investment not accounted for by patents. Difference-in-difference analysis also supports a complementarity between applying to register trademarks and innovative activity, showing higher average R&D expenditure and patenting by first-time trademark filers both before and after initial filing compared to control firms. These initial results warrant further investigation. Still, they provide some preliminary evidence that trademark filings are correlated with firm innovation and that trademark-based metrics may serve to improve measurement of innovation in the economy.

The paper is organized as follows. The next section gives a brief overview of the prior literature leveraging firm and trademark application data predominantly from countries other than the United States. Section 3 provides the theoretical motivation for analyzing the connection between trademarks, applications for trademark registration, and firm characteristics, dynamics, and innovation. Section 4 discusses the data inputs and the algorithm used to match the trademark data with data on firm characteristics. The analysis in Section 5 documents the characteristics of firms that seek trademark registration, and provides a first look at how trademark filing is

[4] See, for instance, Fazio, Guzman, Murray and Stern (2016).

correlated with firm growth and innovation. Section 6 concludes with a discussion of the streams of future research made possible by the trademark-firm linked data.

2 Prior literature

There is a small but growing body of empirical work concerning trademarks. Much of the recent work examining trademark data at the firm level is limited to the European and Australian context. For the United States, most research leverages datasets of publicly-traded firms, such as Compustat, which cover a fraction of those firms seeking trademark registration. The lack of comprehensive data on trademark filing by private firms and small to medium-sized enterprises has been a major impediment to empirical research regarding the use of trademarks and the relationship between trademark filing and firm characteristics, dynamics and innovative activity in the United States.

Schautshick and Greenhalgh (2013) provide a comprehensive survey of the empirical research on trademarks. Several conclusions emerge from the survey. First, there has been considerable growth in trademark application demand since the mid-1970s in Australia, the United Kingdom, and the United States, with qualitatively similar trends in trademark filing growth across these countries between 1975 and 2002.[5] The services sector, as well as deregulated and restructured industries, exhibit the highest rates of growth in trademark filing growth during this period. Interestingly, the surge in trademark applications appears to lead a similar surge in patent filings in developed countries by about ten years. Still, there is no formal econometric analysis that establishes the connection between the timing of patents and trademarks at the firm-level, and whether firm trademark filing leads or lags patent applications is an open question. Second, at the country level, increased demand for product variety and quality appears to drive growth in trademark applications compared to the mere expansion of output. In general, the studies surveyed indicate that trademark filing is correlated with product differentiation, marketing, and innovation. However, the degree of this correlation depends on the industries investigated. Finally, firm-level studies indicate that firms use trademarks to protect identity and reputation and the private value of trademarks to firms is generally positive, but there is no clear conclusion on the social value of trademarks. The latter depends on the trade-off between market efficiency-improving and pro-competitive effects of trademarks and the potential for firms to make inefficient investments to protect reputation and to leverage reputational assets to promote barriers to entry. Further research is needed to assess the relative magnitudes of these different effects.

At the macro-level, Baroncelli, Fink and Javorcik (2005) also document a number of regularities in trademark registrations across countries. Using World Intellectual Property Organization

[5] See Figure 2 in Schautschick and Greenhalgh (2013).

data for a panel of countries over the period 1994-1998, they find evidence that higher development is correlated with a dominance of domestic brands at home and a stronger presence of these brands in foreign markets, as indicated by foreign residents' share of trademark registrations. However, they also note growth in foreign trademark registrations held by entities in by developing countries, potentially reflecting increased exports to markets in more developed countries and the resulting need to protect growing brands. At the sector and industry level, Baroncelli, Fink and Javorcik (2005) observe that most trademark registrations occur in industries characterized as R&D intensive, particularly scientific equipment and pharmaceutical sectors, and advertising-intensive, such as clothing, footwear, and food products. Business services also exhibit higher trademark registration intensity – a finding echoed by Millot (2011) for Germany and France.

Multiple studies document the relatively rapid growth in service marks between 1980s and 2000s in developed countries – Greenhalgh, Longland, and Bosworth (2003) for the U.K., Jensen and Webster (2004) for Australia, and Graham, Hancock, Marco, and Myers (2016) for the United States. In particular, the latter study finds that service marks filings grew by nearly 50% between 1998 and 2000 in the United States, potentially as a result of the dot.com boom of the late 1990s.[6] These patterns reflect the growing importance of the service sector in developed economies and potentially denote a rise in service-related innovations the value of which is captured through trademarks.

A handful of studies leverage firm-trademark matched datasets to examine the connection between firm characteristics and trademark filing or registration activity. Greenhalgh, Longland, and Bosworth (2003) use panel data on medium and large manufacturing U.K. firms and find an inverse relationship between trademark filing intensity and firm size. Smaller firms, as measured either by employment or sales, exhibit higher trademark filing intensity. However, the panel consists mainly of large, publicly-listed firms with many subsidiaries, which calls into question the applicability of results to the broader population of firms relying on trademarks.

Greenhalgh and Rogers (2008) use data on both manufacturing and service firms in the U.K. to investigate the role of firm characteristics on activities related to intellectual property, including trademark filing. They find that, while intellectual property assets are not always monotonically related to firm growth, size is nevertheless a strong predictor of whether a firm applies for a patent and/or seeks trademark registration. Their study also indicates diminishing returns to firm size in terms of such activity. In both services and manufacturing, patent and trademark filing intensity declines as firms get larger. The results of both Greenhalgh, Longland, and Bosworth (2003) and Greenhalgh and Rogers (2008) denote the significant relationship between trademark filing intensity and firm size in the U.K., a finding replicated by Jensen and

[6] See Figure 16 in Graham, Hancock, Marco and Myers (2016).

Webster (2006) for Australian firms.

Sandner (2009) provides a detailed investigation of companies' trademark portfolios using the world's largest publicly-traded companies – those included in the financial databases Reuters and Compustat. The study identifies 4,085 companies that satisfy the selection criterion in their latest income statement.[7] Results indicate that companies build trademark portfolios to deliberately protect the company brand. Using trademark applications to infer brand management decisions, the study finds that product introductions prompt varied decisions regarding whether to extend existing or devise novel trademarks to cover new products. Thus, trademark filings can reflect both the creation of new brands and the expansion of existing brands to encompass new products.

Sandner and Block (2011) use data from various countries, including the United States, to assess the market value of trademarks. They find a positive effect of trademark registrations on firm value, controlling for patenting and R&D activity. However, their study is limited to Community Trademarks registered by the Office for Harmonization in the Internal Market (now the European Intellectual Property Office or EUIPO) and publicly-traded firms from Compustat. The final dataset consists of a relatively small set of 1,216 large, publicly-traded firms, and the results, like those of most prior studies at the firm-level, provide little information on trademark registrations held by private firms.

A few studies have focused specifically on trademark filings by small to medium-sized firms. Rogers et al. (2007) build a database that spans the entire set of U.K. firms for the period 2001-2005. The database contains millions of small to medium-sized firms matched with trademark applications and substantially expands on the scope of the studies discussed earlier, even though the time period covered is relatively short. One main conclusion that emerges from this study is that smaller firms are more intellectual property intensive, tending to have higher volumes of patent and trademark applications relative to their assets, compared to larger firms.

To examine such firms' motivation to seek federal U.S. trademark registration, Block, Fisch, Hahn, and Sandner (2015) use an online survey of 600 small and medium trademark applicants in internet and technology sectors selected from CrunchBase (formerly TechCrunch), a crowd-sourced database of U.S. firms. They use factor analysis to establish three distinct motivations for seeking trademark registration – protection, marketing, and exchange. They then use cluster analysis to build a typology of firms based on trademark motivations, resulting in four clusters – trademark skeptics, marketing-focused trademark users, marketing plus protection-focused trademark users, and trademark advocates.[8] This study demonstrates that there may be signif-

[7] Companies with revenues of 400 million Euros or more (as of the time of the analysis).

[8] Trademark advocates value all trademark motives highly, while the marketing-focus group values trademarks for marketing purposes but do not report protection or exchange as motivation for filing. The marketing plus protection-focused group is the largest, comprising one-third of the firms, and ranks both marketing and protection motives highly.

icant heterogeneity in firm motivations for seeking trademark registration. In a related paper using a comparably sized sample of small and medium firms in Belgium, Flikkema, De Man, and Castaldi (2014) investigate how companies use and interpret trademarks. Their analysis reveals that 60% of recently registered trademarks indicate innovative activity, mainly in the form of product or service innovation. These results encourage further scrutiny into trademark filings by small and medium firm, especially in the U.S. where research has been largely limited to large, publicly-traded firms.

Recent work has also utilized the USPTO Case Files Dataset in conjunction with other datasets in the U.S. to study incentives to use and protect trademark and assess the value of trademarks for firms. Aurora, Bei, and Cohen (2016) explore the incentives to use and protect trademarks for firms in the U.S. by bringing together survey data from The Division of Innovative Labor on firms' new product development activities and industry-level data from the NETS database to understand the connection between trademarks, competition and first-mover advantages. Heath and Mace (2017) offer evidence, using the Federal Trademark Dilution Act and its subsequent revision, that trademark protection through registration has economically significant effects in the case of publicly-traded firms in Compustat. These studies, nevertheless, do not provide a comprehensive analysis of trademark filing propensity for all firms in the United States.

The existing literature, summarized so far, highlights the need for comprehensive longitudinally linked firm-level trademark data in the United States, especially for privately-held firms. The rest of the paper describes how this type of dataset is constructed by combining trademark-related information from the USPTO Case Files Dataset with data on firms, public and private, available at the U.S. Census Bureau. It then demonstrates how the new dataset can be used to better understand the connection between trademark filing and firm attributes, dynamics, and innovation.

3 Theoretical motivation

The theoretical literature sets forth a variety of ways trademarks can be related to firm outcomes and performance.[9] A fundamental function of trademarks highlighted in theory is that of an information signal to promote market efficiency and reduce consumer search costs, especially for experience goods. As source-identifying devices, trademarks convey information regarding the unobserved attributes of a firm and its products (e.g. quality) and therefore reduce information asymmetry and consumer search costs, particularly in markets where the attributes of goods or services are not readily discernible. Firms are incentivized to invest in goodwill to reap rewards

[9] See, e.g., Landes and Posner (1987, 1988, 2003), and Economides (1988), for reviews and assessment of the theory of trademarks.

from the reputational value exemplified by the trademark. Thus, trademarks with positive reputational value can facilitate customer acquisition, generate customer loyalty, and facilitate scale and scope expansion. As a result, trademarks can reduce the price elasticity of demand, allow firms to maintain higher prices and facilitate investment into not only further reputation-building activities but also R&D and innovation. Additionally, by protecting a firm's intangible assets and stock of goodwill, trademarks insulate firms from competition and infringement of their products or services.

In view of the various theoretical roles of trademarks summarized above, one expects both strong selection and treatment effects associated with application for trademark registration. Firms for which reputational assets would yield higher returns are more likely to select into trademark registration. A more productive firm with better growth prospects, a firm that can commit to high quality, or a firm with a large customer base stands to gain more from trademark registration because the benefits accrue from a larger current and future stock of goodwill. Thus, firms that seek to register a trademark for the first time may be those that are larger, more productive, have better product quality, and experience faster growth. In particular, young firms with a innovative product or service expected to yield a large future profit stream may seek trademark registration with a higher propensity.

Trademarks can also induce potentially large treatment effects. Trademarks can contribute to the firm's ability to expand into other product types and new markets, based on the established brand name, goodwill, and loyalty. Trademarks can also prevent the infringement of a firm's brand and the erosion of its stock of goodwill. In markets with relatively homogeneous goods, trademarks can serve a role akin to product differentiation based on quality and reduce the intensity of price competition. This reduction can lead to higher markups and growth for firms that can successfully differentiate themselves using trademarks. All of these effects suggest that the post-trademark filing dynamics of a firm can differ from both the pre-trademark filing dynamics and the dynamics of firms that do not seek to register trademarks at all.

The innovative activity of firms is likely to have a significant impact on both selection and treatment effects associated with trademarks. Firms that engage in R&D and patenting may be more likely to seek trademark registration ex post to appropriate greater returns from their innovations. For instance, a firm that invests in product and process R&D is more likely to generate higher quality products, sustain lower costs, or induce more demand and customer loyalty, and hence, build a larger stock of goodwill. Such firms have greater incentive to protect accumulated and anticipated goodwill with a trademark registration. Therefore, when innovating firms select into trademark use, an application for trademark registration would follow R&D and patenting as a lagging indicator of innovative activity.

Trademark filing may also directly reflect innovative activity not captured by R&D or patents.

Many service innovations, or innovations in information and knowledge-intensive industries, may be better protected by trademarks, rather than patents. More generally, where innovations are not patent eligible subject matter or were developed with informal protection mechanisms, firms may be more likely to seek trademark registration to protect against imitation and secure the firm's current and future stock of goodwill. To the extent that trademark applications reflect such innovations and their associated goodwill, the knowledge of which firms apply for trademarks can enable more accurate identification of the broader population of firms engaging in innovative activity.

Trademarks may also enhance firms' innovation activity. A large stock of goodwill, accumulated and protected by trademarks, can increase a firm's incentives for innovation. A firm with many loyal customers can benefit more from cost-reducing R&D, since reductions would spread over a larger customer base. Similarly, product introductions can be more valuable and involve less risk of imitation for firms with a trademark, as new products can readily enjoy the existing protection and established goodwill of the firm's trademark. Furthermore, where strong goodwill enhances market power, firms with trademarks may be more likely to invest in exploratory R&D. These considerations suggest that firms may engage in R&D and patenting more intensely after they secure a trademark registration. In this sense, trademark filings may also be leading indicators of innovative activity.

In summary, theory suggests that there may be strong selection and treatment effects associated with a trademark application. Both the pre- and post-trademark evolution of firms with a trademark application may differ significantly from that of firms that do not seek trademark registration. Furthermore, innovative activity captured by R&D and patenting can be strongly correlated with trademark filing. These two types of activities can complement each other, leading to high firm-level correlation between trademark application, R&D, and patenting. The relative timing of innovation and trademark filing over the firm life-cycle may be informative in understanding whether trademark applications can serve as precursors of innovation. Empirical analysis in the following sections will explore some of the selection and treatment effects associated with trademark filings suggested by the theory of trademarks. It will also examine the connection between trademark filings and innovative activity, as measured by R&D and patents, at the firm level.

4 Data

This section describes the datasets used to link trademark application filing information with longitudinal firm data. It provides an outline of the methodology for matching trademark data to firms. Because the trademark data consists of many separate files and a large number of variables,

it is important to develop a strategy for using all the relevant information from trademark applications by firms to facilitate the matching process.

4.1 Data on trademarks

The data on trademarks comes from the USPTO Trademark Case Files Dataset (TCFD). This dataset was constructed by economists at the USPTO from trademark case files made available by the USPTO on the `data.gov` website. The case files were organized and streamlined to form several electronic files that can be readily used by researchers to conduct large-scale analysis. The accompanying paper, Graham, Hancock, Marco, and Myers (2016), provides an excellent account of how the TCFD was constructed and a first look at some of the patterns of trademark filing activity that emerge from the data. The TCFD contains detailed information on USPTO trademarks applications and registrations for the period 1870-2015. The information on trademarks includes, but is not limited to, ownership, assignment, prosecution events, classification, and renewal history.

In the United States, trademark registrations are subject to a use requirement, which obligates the owner to use the mark on goods or in connection with services in order to establish and maintain trademark rights. The use requirement derives from American common law and subsequent codification in federal statute.[10] An entity establishes and can enforce common law trademark rights solely by using a mark in commerce. A federal U.S. trademark Principal Register registration confers benefits beyond common law, specifically national-scope rights, prima facie evidence of ownership, and recordation with the U.S. Customs and Border Protection for preventing the importation of infringing goods.

The TCFD captures only information on entities that seek a federal U.S. trademark registration. It does not capture the population of firms that rely solely on common law trademark rights. This is an important distinction because the selection and treatment effects considered here are limited to those associated with filing for federal trademark registration at the USPTO. Results may not be applicable to the broader set of firms with only common law trademark rights.

[10] Under American common law, a trademark owner has the exclusive right to prevent unauthorized third parties from using the same or similar mark on goods and services where such use would likely cause confusion among consumers as to the source of the goods and services offered under the mark. An entity may establish trademark rights solely by using a distinctive mark on the goods or in connection with the services. Registration at the state or federal level provides additional benefits but is not necessary for an entity to create and enforce common law trademark rights. The Lanham Act of 1946 ("Trademark Act") established the modern U.S. federal trademark registration system, providing for protection of trademarks used in commerce and registered with the USPTO. 15 U.S.C. §1051 et seq.

To file a U.S. trademark application, an applicant must specify the goods and services on and for which she uses or intends to use the trademark. The identified goods and services define the scope of trademark protection covered by the registration and generally cannot be overly broad.[11] Still, even within the same class, there can be considerable variation in the specificity of the goods and services listed and, thereby, the breadth of trademark protection.

An applicant can apply to register a trademark already in commercial use or for which she has a bona fide intent to use the trademark on the identified goods or in connection with the identified services.[12] However, such so-called "intent-to-use applications" can only be registered after the owner uses the trademark in the ordinary course of trade in commerce and provides a declaration and evidence supporting such use to USPTO.[13] Filing for a U.S. trademark registration costs a relatively small fee per class.[14] During substantive examination, the USPTO determines if the applied-for trademark is legally protectable and that there is no "likelihood of confusion" with a previously registered trademark owned by another party.[15] If the examining attorney determines the applied-for mark is registrable, the USPTO publishes the trademark for a limited opposition period, during which time third parties may file a formal opposition to the registration. Oppositions are fairly rare, instituted in less than 3% of published applications (Graham et al., 2013). Unopposed applications for trademarks already in use are issued a U.S. trademark registration. Allowed intent-to-use applications must first establish commercial use before the

[11] The U.S. adopted the International Classification of Goods and Services under the Nice Agreement (the so-called "Nice Classification") in September 1, 1973. Prior to that date, the USPTO used a U.S. trademark classification system. Our match is restricted to trademarks registered under the Nice Classification system.

[12] To file based on use in commerce under Trademark Act §1(a), the owner must submit a declaration stating that the mark is used in the ordinary course of trade in commerce that Congress can regulate, i.e., interstate commerce or commerce between the U.S. and foreign nations, as of the filing date. See TMEP §901.03. To file based on intent-to-use under §1(b), the applicant must have a bona fide intention to use the mark in commerce on the goods and services listed in the application in the near future. See TMEP §1101.

[13] Intent-to-use applications became available to applicants in November 1989 as a result of the Trademark Law Revision Act of 1988. A small, but growing, minority of applications are filed with the USPTO based on a foreign application or registration for the same trademark or an extension of an international registration to the U.S. Authorized by international treaties, such applications can be registered prior to the trademark being used in U.S. commerce, however, generally only applicants with a foreign "country of origin" can obtain such U.S. trademark registration. Owners with foreign addresses are excluded from the match with U.S. firm data.

[14] For most of the time period for which the matched data cover, the per class filing fee ranged from $175 to $375 for a paper filing and $325 to $335 for an electronic filing.

[15] An applied-for trademark can be refused as not registrable if, among other possible grounds, it is generic or merely descriptive, geographic, a surname, deceptive, a municipal, state, national, or foreign flag or insignia, or the name, likeness, or signature of a living person used without their consent. 15 U.S.C. §1052. See TMEP §1200. Examining attorneys search existing registrations and pending applications for similar trademarks and assesses whether use of the applicant's trademark on the identified goods or in connection with the identified services is likely to cause confusion among consumers. 15 U.S. C. §1052(d). See TMEP §1207.

applied for mark can be registered.

A trademark owner can renew a U.S. trademark registration indefinitely so long as the trademark is used on the listed goods or in connection with the listed services. The owner must provide proof of continued use and pay prescribed fees to the USPTO six years after registration and at each ten-year renewal event.[16] Failure to do so results in the registration being canceled.

Data coverage in the TCFD varies over time. Graham, Hancock, Marco, and Myers (2016) indicate that there is little coverage of classification, prosecution events, and owner records for trademark registrations issued before 1962.[17] The coverage improves for registrations issued during the period 1962-1977, and there is substantial improvement in coverage for filings and registrations after 1977. Coverage becomes nearly 100% after 1982. Key data items are populated at a rate of 89% or more for the period 1977-2015. Fortunately, as discussed below, firm-level longitudinal micro data is available in the U.S. Census Bureau starting in 1976. Thus, much of the well-covered 1977-2015 period in the TCFD coincides with the coverage of the data on firm characteristics. However, left censoring of the firm data at 1976 and relatively lower coverage of the trademark data before 1977 implies that firms born before 1976 may not match with any trademark data, even if they applied for or registered a trademark before 1977. Likewise, for firms born before 1977, there is no way to assess whether a matched trademark filing reflects the firm's first trademark application - a key trademark-related event in the life-cycle of firms. Therefore, the primary focus of the empirical analysis will be on firms that were born in or after 1977.

4.2 Data on firms

The trademark data is matched with the U.S. Census Bureau's Business Register (BR), which contains administrative data for the universe of non-farm businesses in the United States. The BR is also the sampling frame for the Census Bureau's economic surveys. It contains information on a firm's employment, payroll, and revenues, as well as geography and industry classification. Analysis of firm trademark activity requires longitudinal data to track firms over time and identify

[16] In the sixth year after the registration date, the trademark owner must maintain the registration by filing an affidavit or declaration of continued use, provide specimen(s) depicting use in U.S. commerce for the listed goods and services and pay prescribed fees. 15 U.S.C. §§1058(a)(1). See TMEP §1604. Ten-year renewal terms were instituted for registrations issued on or after November 16, 1989. Registrations issued prior to that date had twenty-year terms until the first renewal event following that date. Thus, all live registrations are subject to ten-year terms as of November 16, 2009. Registrants must pay separate maintenance and renewal fees for each class in the registration. For most of the time period for which the matched data cover, the fee for (paper or electronic) filing an affidavit or declaration of use is $100 per class and the fee for (paper or electronic) filing a renewal application ranged from $300 to $400 per class.

[17] See Table 1 in Graham, Hancock, Marco, and Myers (2016).

when in their life-cycle such activity occurs. The Longitudinal Business Database (LBD) provides a longitudinally linked version of the BR at the establishment level for the period 1976-2014. The LBD also enables identification of entry and exit of firms and establishments. Since the LBD coverage starts in 1976, there is no age information for firms established in that year. The empirical analysis is restricted to firms born in or after 1977 to avoid this censoring in age and ensure accurate identification of the first occurrence of a trademark filing in a firm's life-cycle.

The primary measures of firm size used in this paper are employment and revenue. Prior work, detailed in Haltiwanger, Jarmin, Kulick, and Miranda (2016), has linked observations in the BR to construct a longitudinal firm revenue database. The analysis here uses this database to examine the connection between firm revenue growth and trademark filing.

In addition to the BR, the matching process utilizes the Integrated Longitudinal Business Database (ILBD). Even after matching to the BR, there are many trademark applications in the TCFD left unmatched with a firm. Many of these trademarks may be owned by non-employer businesses that do not appear in the BR. The ILBD contains individuals' income tax records including Schedule C earnings from an individually-operated business or sole proprietorship.[18] The ILBD non-employer universe is used to identify matches for those trademark applications that were not matched to the employer universe in the BR. These non-employer trademark links are used primarily to better understand the accuracy of the matching process. Future work will leverage these links to examine the dynamics of non-employer firms seeking federal trademark registration.

4.3 Matching process

Because there are no disambiguated identifiers, such as an Employer Identification Number (EIN), shared between the TCFD and the BR, name and address matching techniques must be applied to combine the two datasets.[19] The TCFD contains over 7.2 million trademark applications and 17.4 million ownership records.[20] The universe of TCFD records are filtered to include only the applications filed in or after 1976 (the beginning of the BR data) and exclude foreign businesses, federal and state government entities, and all individuals. After imposing these restrictions, there are over 5 million unique trademark records at risk for a match to the BR and LBD.

The first step of the matching strategy to link the TCFD and BR is to extract all unique combinations of name and address information from the TCFD. Matching algorithms then clean

[18] See Davis et al. (2007) and Haltiwanger and Jarmin (2007) for details on the construction of the ILBD.

[19] For a detailed description of the matching methodology see Appendix A.

[20] Ownership records are captured in the TFCD at key points in the trademark's life-cycle (e.g., filing, publication, registration, etc.) even if there is no change in ownership. This inflates the number of ownership records in the TFCD beyond the number of unique trademark application-owner pairs.

and standardize the name and address fields in both the TCFD and the BR.[21] Once standardized, an initial subset of potentially matching records is identified based on weak match criteria applied to names only. From this subset, matches of different quality are extracted using various combinations of fuzzy and exact name and address matching. A string comparator that captures the similarity of text across fields is used to further refine and subset matches. More than 80% of matches rely on the business name and three address (street, city, and state) fields – a relatively strict criteria that tends to yield higher quality matches. Results from the BR match are then integrated with the LBD. Information from the LBD is leveraged to further refine matches.

A trademark application filing can predate a firm's entry to the BR, as some firms may apply for a trademark registration even before they become employers (pay their first wage or hire their first employee). Thus, trademark applications that do not match to the BR are matched to the ILBD, the universe of non-employer businesses. For this study, matches to non-employer businesses are used primarily to better understand the quality of matches to the employer universe. Future work will investigate the growth and transition dynamics of non-employer businesses seeking trademark registration.

Ultimately, 83% of trademark records match to the LBD and 2.4% to the ILBD. The LBD match rate declines over time, falling from 89% in the 1980-1989 registration year period to 73% for the 2010-2014 period. However, the matches become less ambiguous over time. The share of unique matches rises from 59% in the 1980-1989 registration year period to 88% for 2010-2014. The declining match rate is primarily driven by businesses identified as Limited Liability Companies (LLCs) in the TCFD. LLCs are one of the fastest growing business types in the trademark applicant universe and in the general population of firms. To explore the declining match rate, several hypotheses are investigated, including deteriorating string quality, a compositional shift in trademark filings towards non-employers, and the growth of informal business. As detailed in the Data Construction appendix, there is no obvious culprit that can explain why a declining share of trademark owner records in the TCFD match to the LBD. A random sample of matches are classified by clerical review as either true or false positives. These classifications are then used to calculate the precision of the TCFD-LBD matches. The precision of matches is over 90% and remains stable over time aside from a slight decline to 87% in the 2010-2014 period.

5 Analysis

This section provides a first look at the characteristics of firms seeking federal trademark registration in the United States based on the constructed data. For the purposes of this paper, a

[21] Common strings, such as "LLC" and "LTD", and punctuation, such as "." and "@", are removed.

"trademark filing firm" is defined as one that has filed an application for a trademark at some point in its life-cycle. In view of the theoretical motivation, the main goal of the empirical analysis is to understand both selection and treatment effects – how trademark filing firms differ from those that do not file trademark applications, both before and after their first trademark filing. At the firm-level, first trademark application is identified by the first ever filing of an application to register a trademark with the USPTO. The analysis therefore focuses on selection and treatment effects of trademark filing activity via the federal trademark system, and non-trademark filing firms comprise firms without any trademark applications, regardless of whether they do or do no own trademarks under common law. While it would be ideal to separate non-trademark applicants into two subsets, those that own common law trademarks and those that do not, the lack of comprehensive data on common law trademark rights-holders prevents such differentiation. However, since the dynamics of firms with common law trademark rights would tend to be more like those of formal trademark filing firms, the inclusion of the former in the non-trademark filing group would tend to reduce, rather than inflate, selection and treatment effect estimates.

It should also be noted that not all trademark applications mature to registration. Applications can be abandoned during substantive examination or following allowance for failure to establish commercial use of the applied for mark or as a result of third-party opposition proceeding. The analysis presented here focuses on trademark application filing because it is the first indication of a firm's intent to register a trademark. It endeavors to fill a gap in the prior literature regarding which firms seek trademark registration and when in their life-cycle they enter into this activity. However, because legal benefits accrue from trademark registration, not application filing, treatment effects would tend to be larger for firms whose applications mature to registration. Thus, including all trademark filing firms, regardless of whether an application proceeds to registration, would yield more conservative estimated treatment effects. Subsequent research will also consider the treatment effects of registration alone.

For the empirical analysis, only firms born in or after 1976 are considered. As discussed before, this restriction ensures that left-censoring of firm age does not affect results, but it also renders the age distribution of firms to be heavily skewed towards young firms in the earlier part of the sample period. The age distribution gradually evolves to be more representative over time. In particular, in 1977 the only firms in the sample will be new firms born in that year. However, for any given year $t > 1977$, firm age will range from zero (firms born in year t) to $t - 1977$ years (firms that were born in 1977 and survived till at least year t). Some of the empirical analysis considers only the later years of the sample period when the firm age distribution is more representative. Specifically, the results for $t = 1997$ are presented whenever the analysis demands a representative firm age distribution.

The analysis focuses on the first ever filing to register a trademark by a firm, regardless of

whether the trademark is eventually registered or not. This critical event denotes when in their life-cycle firms select into the population of firms that apply to register a trademark. In principle, one can use the filing date variable in the TCFD associated with each trademark application to identify the timing of the firm's first trademark activity. However, the process is a bit more complicated, because there can be several applications for the same trademark corresponding to use in different classes or on different products within the same class. For instance, a firm may initially apply to register a trademark for computer hardware services, but may later extend the same trademark to another product offering, such as customized computer software. For cases where there is more than one application for a trademark or multiple trademarks, the minimum filing date across the firm's portfolio of trademark applications is used to identify the first ever trademark filing.

5.1 Trends in trademark filing by broad sectors of the economy

To paint a broad picture of trademark filing activity in the United States, consider first some general patterns of trademark applications across sectors of the economy. Figure 1 shows the distribution of firms that file to register a trademark for the first time by sector based on their Standard Industrial Classification. Note that the number of first-time trademark filing firms grows initially for all sectors starting with the beginning of the sample through the 1990s. This trend is, in part, driven by the fact that only new and relatively young firms are present in the sample for the earlier years, as discussed earlier. Since younger firms may have more incentive to apply for trademark registration relative to older counterparts, the number of first-time trademark filers increase as the stock of young firms initially expands and dominates the sample age composition. Later in the sample period, the distribution of firm age approaches a more representative one, and the number of first-time trademark filing firms becomes relatively more stable in many sectors.

There is a notable expansion of first-time trademark filing by professional services firms, starting with the 1990s, when the firm age distribution becomes relatively more representative, and extending through the mid-2000s. Similar growth in first-time filing is evident for wholesale firms and, to a limited extent, for retail firms. In contrast, first-time trademark filing by manufacturing firms has somewhat declined during the same period, likely reflecting the decline of manufacturing in the U.S. that has accelerated especially in the 2000s. Surprisingly, there is more entry into trademark filing by wholesale firms compared to retail firms over the sample period, possibly indicating greater importance of goodwill in business-to-business transactions. Throughout the sample period, first-time trademark filers in the "Other Firms" category are both the largest and fastest growing, which is expected given that this category lumps together

industries that are likely to be trademark intensive, such as information, education and health care, entertainment, accommodation and food and other services.[22]

The dot.com era spike in trademark applications is also clearly visible before 2000, mainly for professional services firms and "Other Firms".[23] This spike is potentially related to the proliferation of internet-based commerce and a need to identify and protect brands in cyberspace. As the 2001 recession hits, first-time trademark filing declines sharply. Thereafter, entry by new trademark filers increases across sectors, except manufacturing, until the Great Recession in 2007. Overall, first-time trademark filing appears to be pro-cyclical, at least in recent decades. By the end of the sample in 2013, the largest number of first-time filers are in the "Other Firms" category, followed by professional services, wholesale, and manufacturing. Among all sectors, retail has the fewest number of first-time trademark filers as of 2013.

5.2 Who trademarks, when, and how much?

This section examines the patterns of trademark filing by firm size and age, the two key conditioning variables frequently used in the firm dynamics literature. The number of employees and the age of firms when they first file a trademark application are important in understanding the potential selection effects of trademarks. This section also explores the intensity of trademark filing, as measured by the number of trademark applications per firm, and firm size and age.

5.2.1 Firm size, age, and trademark activity

Table 1 presents, by year, the average size (employment) and age of firms that filed for a trademark registration for the first time during the period 1997-2013 (labeled as "Firms with first TM filing"). It also contains, for comparison, the same statistics for firms that have not filed to register a trademark as of the specified year (labeled as "Firms with no TM filing"), have at least one trademark filing up to that year (labeled as "Firms with a TM filing"), and filed for a new trademark registration in that year (labeled as "Firms with current year TM filing").

The main message from Table 1 is that first-time trademark filing firms are young but also large relative to firms that do not seek trademark registration. The average firm age at the time of the first trademark filing ranges from 4 to 6 for most years included in Table 1, indicating that many firms that seek trademark registration for the first time do so relatively early in their life-cycles. The average age of first-time trademark filing firms increases over time. However, this pattern is evident in each category in Table 1, reflecting, in part, the overall aging of the

[22] "Other Firms" category include the firms classified in the industries outside of manufacturing, retail, wholesale and professional services. See Table 2 for a list of all industries.

[23] These trends are also seen in High Tech industries, where young firm activity surged in the 1990s and then collapsed after 2001. See Goldschlag and Miranda (2015) for details.

U.S. firm population as a result of the persistent decline in new firm entry – a trend extensively documented in recent research.[24] The rise in mean age over time is also driven by the fact that firms in the analysis get older as one moves further away from the initial year of the sample (1997).

The first-time trademark filers are also relatively large. For instance, in 1997, firms that first filed to register a trademark had an average of roughly 76 employees, nearly 8 times the average employment of firms that had no trademark applications. This gap narrows in subsequent years, mainly because the average size of first-time trademark filing firms declines while that of non-trademark filing firms is largely constant. Still, in 2013, the average employment of first-time trademark filers was roughly 58, about 6 times that of firms with no trademark filings.

Table 1 also shows that firms that have filed for at least one trademark ("Firms with a TM filing") are larger and older than the firms that do not apply for trademark registration. Likewise, firms that apply to register additional trademarks in any given year ("Firms with current year TM filing") are much larger than non-trademark filing firms, but also tend to outsize firms that have at least one trademark application. Interestingly, these firms that continue to build their trademark portfolios tend to be younger than firms with some trademark filing activity. One interpretation of this result is that firms that continue to seek trademark registration are mostly large and successful companies that expand their scale and scope by introducing new products and services under the brand name and goodwill already established.

The patterns in Table 1 are also apparent by sector. Table 2 presents the average firm size and age for the panel of trademark and non-trademark filing firms across all years in the 1997-2013 period. In each sector, firms that apply to register a trademark for the first time are, on average, much larger than firms that do not seek trademark registration. The difference in average size is the largest in agriculture, mining, and utilities, where first-time trademark filing firms maintain 16 times more employees, on average. The smallest difference is in retail, where the average employment of first-time filers is more than double that of non-trademark filing firms. Sector-level differences in average firm age also mirror the pattern found for the general population of firms in Table 1. Across sectors, first-time trademark filing firms are, on average, 2.5 years younger than the firms without trademark applications. The largest age disparity is in manufacturing, where first-time filers are nearly 5 years younger, on average, than firms that do not apply for trademark registration. In construction, transportation, and warehousing, however, the difference in average age between first-time filers and non-trademark filing firms is only 0.5 years.

Across sectors, firms that have at least one trademark application and those that apply to register additional trademarks are, on average, 10 and 25 times larger, respectively, than firms

[24] See, e.g., Decker, Haltiwanger, Jarmin and Miranda (2016).

that do not seek trademark registration. Again, the largest differences in average employment are in agriculture, mining, and utilities as well as finance and real estate.

Figure 2 plots the relationship between the propensity of a firm having applied to register a trademark and its size measured by employment. The likelihood of a trademark filing increases with size for all sectors, though the patterns of growth differ. The probability of having a trademark filing increases much faster with employment for smaller firms in wholesale, manufacturing, and professional services. For example, the probability that a firm with roughly 20 employees ($\simeq 3$ in log scale) in one of these sectors files to register a trademark is 0.20 to 0.30. This probability increases to around 0.40 at 55 employees ($\simeq 4$ in log scale). For retail, on the other hand, the likelihood of a trademark filing is relatively low until approximately 150 employees ($\simeq 5$ in log scale), after which it grows sharply and catches up to the other sectors. For any given firm size, the probability of a trademark filing is generally highest in manufacturing and wholesale sectors, though professional services is a close third and retail is roughly equivalent for very large firms. In general, Figure 2 reinforces the results in Tables 1 and 2 and indicates there is a strong connection between firm size and trademark filing.

The relationship between firm age and propensity to apply to register a trademark, plotted in Figure 3, is not as strong. While the probability that a firm has a trademark filing increases with age in general, the rate of growth is much less pronounced compared to size. In addition, the relationship is not necessarily monotonic.

Across sectors, the steepest increase in the probability of filing to register a trademark generally occurs between age zero (new firms) and age five. As firms age, there is little to no, or even negative, growth in the probability of trademark filing. The most sustained growth is in manufacturing, where the probability rises from roughly 0.08 to 0.20 between age zero and age 36 (the oldest firms in the sample). Combined with Figure 2, the patterns in Figure 3 suggest that firm size is a more critical correlate of trademark application than firm age and that firms who do not seek trademark registration in the early years of their life-cycle are unlikely to do so unless they experience employment growth. Thus, the relationship between firm growth and trademark filing transcends mere firm experience and survival, hinting that the decision to apply for a trademark registration is a strategic one made by the firm.

The analysis so far indicates that first-time trademark filing is concentrated in young and large firms. There appears to be strong selection into trademark filing, at least based on two key observable firm characteristics, size and age. This finding is consistent with the theoretical view that firms that are successful and grow in size early in their life-cycles have greater incentive to formally protect their goodwill through registered trademarks. In addition, the fact that trademark filers are much larger and older than non-trademark filing firms suggest that trademark filing firms continue to do perform better after initial trademark application, pointing to potential

treatment effects of trademarks that will be explored further in subsequent sections.

5.2.2 Firm trademark intensity

The analysis thus far has focused on the characteristics of trademark filing firms without considering how intensely those firms rely on trademark registrations to protect their goodwill. One measure of trademark intensity is the number of trademarks applications filed by a firm, akin to the number of patent applications or products per firm. Figure 4 shows the kernel density plot of the count of trademark applications per firm during the sample period. The distribution exhibits the typical features of that of firm-level discrete outcomes: it is highly positively skewed and has a long right tail. Most firms have a small number of trademark filings (less than 3), but there are also many firms with large trademark application portfolios in the right tail of the distribution – some with at least 15 trademark filings. Such large trademark portfolios may result from firms expanding into related and unrelated business lines to take advantage of established goodwill but can also reflect re-branding or other marketing activities. It is important to note that the skewness in firm trademark application intensity is different than that in patent intensity. Patent applications are highly concentrated among a small number of top patenting firms, whereas trademark filings are much more dispersed and the top applicants hold relatively smaller portfolios.

How much has trademark filing intensity changed over time? Do firms own increasingly larger application portfolios, on average? Table 1 documents the average number of trademark filings per firm by year over the period 1997-2013. For the years covered in the table, the average is around 2 trademark applications per firm. While the average is somewhat higher in the earlier years, it appears to be relatively stable, suggesting little change in overall trademark filing intensity over time.

5.3 Trademarks and firm growth

Both theory and the prior descriptive analysis suggest that there are strong selection and treatment effects associated with trademark application filing. This section focuses on the evolution of firms before and after their first trademark filing, without any attempt to identify a causal link between trademark application and firm evolution. The goal is to determine whether trademark filing firms differ from non-trademark filing firms in key outcome measures, both before and after the initial filing. To do so, the analysis first considers the growth trajectory of newly-formed firms that apply to register a trademark in their first year, compared to those that do not. It then uses nearest neighbor propensity score matching to identify a more precise control group for all treated firms (the ones that apply for a trademark registration for the first time) and

more closely examines the treatment effects associated with first-time trademark filing. It should be noted, however, that the matching estimator does not eliminate concerns due to unobserved characteristics of treated firms.

5.3.1 Firm growth before and after trademark filing: new firms

Figure 5 presents the firm size-age profile for all new firms (age zero firms) that apply to register a trademark in their first year versus those that do not seek trademark registration at all. The figure suggests that firms that file in their first year of existence tend to have a very different growth trajectory compared to non-trademark filing firms. Trademark filing new firms also tend to be larger, even in their first year, compared to non-trademark filing counterparts. This result suggests a strong selection into trademark filing at firm birth based on size. Average employment for first-year trademark filers is higher for any given age, and the employment gap between the two groups widens as firms age. Firms that file for a trademark registration upon birth tend to experience a steeper increase in employment in their first year. Average employment more than triples from roughly 2.5 employees in year zero to nearly 10 employees by year one. While both types of firms tend to grow with age, average employment grows much faster for the firms with a trademark application filing compared to those without one.[25]

Note that these trends are not conditional on industry, year of birth, or any other observables for new firms. The size-age profile of new firms, both trademark filing and non-trademark filing, is likely to vary based on such factors. Likewise, selection into a trademark filing is likely to be correlated with various firm characteristics other than size. To more carefully consider the relationship between trademarks and growth, the next section introduces a propensity score matching method applied to control for the effects of some observables at the firm level.

5.3.2 Firm growth before and after trademark filing: propensity score matching

To form a control group for all first-time trademark filing firms (not just the new firms) based on their observable characteristics, a propensity score matching methodology is implemented. The indicator of first-time trademark filing in any given year is modeled as a function of firm size (employment), age, average payroll (payroll per employee), multi-unit status, industry fixed effects and prior year size in a logit framework. The model is estimated for each year separately. The predicted probabilities from the estimated model are then used to attach a propensity score to all firm-year combinations – treated and non-treated. For each treated firm, a matching firm is selected through propensity score nearest-neighbor matching. The control group is further

[25] Note also that Figure 5 is not conditional on survival, so it does reflect the differences in failure rates for firms that apply for a trademark registration versus those that do not.

restricted to matching firms of the same age as the treated firm in the year of first trademark filing. In some cases, this process yields more than one match for each treated firm. The analysis proceeds with weights, when needed, to account for multiple matches.

Figure 6 plots average employment before and after the first trademark filing for firms that apply to register a trademark (treated) and the matching control firms (untreated).[26] The year of the first trademark filing is normalized to $t = 0$, and is indicated by a vertical line in Figure 6. For the two years prior to trademark filing ($t = -1, -2$), the average employment for treated and control firms is relatively similar. Nevertheless, treated firms are somewhat larger on average, including at the year of the trademark filing. Average employment grows for both treated and control firms before filing. However, while average employment increases substantially for the treated firms after filing, there is a much smaller increase for the control group. As a result, the gap in average size between treated and control groups grows considerably. Five years after first filing ($t = 5$), the average employment of the trademark filing firms is nearly twice that of the control group.

For better visual comparison of the trends, in Figure 7 mean employment levels are normalized to one for both groups at $t = -2$, two years before the first trademark filing. The difference in mean employment between the two groups is statistically significant for each period t.[27] Figure 7 makes it clear that average employment is increasing in the year prior to filing for both treated and control groups. After filing, however, there is continuing growth in average employment for treated firms, but much less expansion for the control group. In particular, average employment for the treated firms five years after the first trademark filing ($t = 5$) is about 80% higher than at time of filing. For the control group, mean employment exhibits little growth in the five years following filing.

Figure 8 repeats the exercise in Figure 7 with average firm revenue for treated and control firms. Pre-filing average revenue trends are similar for the two groups but diverge at the year of first filing. The average revenue gap between treated and control groups is fairly constant through two years after filing, but expands considerably thereafter. For treated firms, average revenue increases by about 100% five years after trademark filing. The control group exhibits much more modest growth of about 35% over the same horizon. As in the case of employment, these patterns suggest the presence of potentially large benefits to trademark filing firms in terms of revenue.

Taken together, Figures 6-8 show considerable differences in the growth of employment and

[26] For the control group, the mean for any firm outcome measure is calculated using weights, which are equal to the inverse of the number of control firms corresponding to a given treated firm.

[27] Confidence intervals are not shown because the large number of firms in the sample generate very precise averages with tight confidence intervals. Therefore, the differences are statistically significant for each t.

revenue for trademark filing firms compared to their matched controls for both pre- and post-filing periods. Pre-filing patterns suggest the potential presence of selection based on unobservables not accounted for in propensity score matching, as treated firms tend to be somewhat larger on average than the controls – though the difference is not substantial. Post-filing patterns also indicate the likely presence of trademark treatment effects. Treated and control firms diverge substantially in terms of average employment and revenue after first trademark filing. The gap in mean employment and revenue between the two groups expands noticeably after two years post-filing. This pattern may simply reflect application pendency at USPTO or suggest that some effects of trademark filing are more gradual and take time to emerge.[28]

To understand the connection between firm growth, innovation, and trademark application filing further, the following difference-in-difference regression framework is considered for the treated firms and their matched controls

$$\begin{aligned} \widetilde{Y}_{it} &= \alpha + \beta_F FILE_i + \beta_P POST_FILE_{it} + \beta_{FP} FILE_i \times POST_FILE_{it} \\ &\quad + YEAR_t + INDUSTRY_i + \varepsilon_{it}, \end{aligned} \tag{1}$$

where the treatment status of firm i is indicated by $FILE_i$, and the time-period t after treatment by $POST_FILE_{it}$. The dependent variable $\widetilde{Y}_{it}$ is the inverse hyperbolic sine transformation of the four dependent variables considered

$$\widetilde{Y}_{it} = \ln(Y_{it} + \sqrt{Y_{it}^2 + 1}).$$

This transformation is useful, in particular because there are several cases of no patenting or R&D expenditure in both treated and control groups. The coefficients β_F and β_{FP} measure, respectively, the effects of being in the treated group (trademark filing firms) prior to treatment, and being in the treated group *and* in the treatment period (years after first filing for trademark filing firms).

Table 3 presents the results from the estimation of (1). The first two columns show the estimates for employment and revenue as the dependent variables that measure size. Note two key overall results in Table 3. First, estimates of the β_F coefficient indicate that treated firms are, on average, larger in terms of both employment and revenue than the control group in the period before trademark filing. This finding indicates the likely presence of unobserved characteristics, not controlled for via propensity score matching, that are correlated with trademark filing. Second, estimates for the β_{FP} coefficient indicate sizable treatment effects. In the period following first trademark filing, treated firms have, on average, approximately 34% higher employment

[28] Average total pendency at USPTO for trademark applications is 8 to 12.5 months depending on method of filing. See https://www.uspto.gov/dashboards/trademarks/main.dashxml

and 24% greater revenue compared to the control group.[29] Overall, the results of the regression analysis for employment and revenue suggest that there are significant selection and treatment effects associated with first trademark filing. These effects will be investigated in further detail in future work.[30]

5.4 Trademarks and innovative activity

Next, consider the firm-level connection between different types of innovative activity and trademark applications. The theoretical motivation in Section 3 suggests that investments in goodwill accumulation and innovation can be complements. An implication is that measures of these two types of investments should be correlated to some degree at the firm level. The next section explores the co-presence of trademark applications and innovative investment, as measured by R&D, and innovative output, as measured by patents. It also examines the timing of the first trademark filing relative to that of R&D expenditures and patenting.

The patent data is derived from patent-firm linked data from the U.S. Census Bureau, which combines the Longitudinal Business Database with the USPTO's patent database.[31] The firm-level R&D data is sourced from the Standard Industrial Research & Development Survey (SIRD) and Business R&D and Innovation Survey (BRDIS), conducted by the U.S. Census Bureau for the National Science Foundation.[32] While the patent and trademark data pertains to the entire set of firms observed in the LBD, R&D expenditure data is only available for firms sampled in the SIRD and BRDIS. To analyze the trademark filing and patent application activity by R&D performing firms, the sample is therefore restricted to those firms that reported some or no R&D activity in the SIRD and BRDIS. While the survey, combined with the survey weights, is intended to be nationally representative, the raw (unweighted) firm counts, which tend to skew heavily towards R&D performing firms, are used.

[29] Note that (1) implies that the percent change in $(Y_{it} + \sqrt{Y_{it}^2 + 1})$ for the treated group in the treatment period can be estimated as $100\left(\exp(\widehat{\beta}_{FP}) - 1\right)$. For Y_{it} not too small, the last estimate also approximates the percent change in Y_{it} due to a change in $FILE_i \times POST_FILE_{it}$ from 0 to 1, because $Y_{it} + \sqrt{Y_{it}^2 + 1} \simeq 2Y_{it}$ – for Y_{it} not too small.

[30] In particular, more stringent matching processes will be explored to understand whether the control group can match the treatment group better during the pre-treatment period. This exercise will further clarify whether the differences in the pre-treatment period between the treated and control groups are due to unobserved characteristics of the treated firms or can be eliminated with further refinement of the matching process.

[31] See Graham, Grim, Islam, Marco, and Miranda (2015). Their work has been extended to include patenting activity by firms from 2011 onwards.

[32] See Foster, Grim, and Zolas (2016) for more details on this survey and the characteristics of R&D performing firms.

5.4.1 Firm-level correlation and relative timing of R&D, patents, and trademark filings

Table 4 presents information on the co-presence of innovative activity among trademark filing firms. It shows the fraction of firms with patent applications and R&D activity for firms who have filed for a trademark registration and all firms in the U.S.. The share of trademark filers with patenting and/or R&D activity remains relatively small (between 8-9% of trademark filing firms own at least 1 patent application and 3-7% perform R&D). However, Table 4 indicates that trademark filing firms are 15 to 20 times more likely to file for a patent or perform R&D, compared to a typical U.S. firm. From the perspective of better measuring firm innovation, this result is encouraging. To the extent that trademark data captures forms of innovation not typically accounted for by patents, this finding suggests that broadening the definition of innovating firms to include trademark filing firms can enhance the identification of the innovative segment of the firm population. On the other hand, if most trademarks are used merely to differentiate largely homogeneous products rather than introduce true product or process innovations, trademark filing firms may contribute little to the understanding of innovative activity by firms.

The picture is very different when considering the co-presence of trademarks filings among patent applicants and R&D active firms. Among the firms which have filed for a patent or performed R&D, Table 5 suggests that the majority have applied for at least one trademark registration, with the rate steadily rising over time. Among patent filing firms, 55% had applied for at least one trademark registration in 1997, with this figure rising to nearly 72% in 2011. Among R&D performing firms, 52% had filed for at least one trademark registration in 1997, and as high as 68% had done so in 2011. More interesting, however is the change in the proportion of R&D performing firms seeking patents versus trademark registrations. In 1997, more than 61% of R&D performing firms filed for at least one patent (versus the 52% that applied for at least one trademark registration). By 2011, the balance of innovative output changed, where only 50% of R&D performing firms filed for at least one patent (versus the 68% that applied for at least one trademark registration). From an innovation measurement perspective, this reversal in output among R&D performing firms supports the notion that, for at least a subset of firms, trademarks may capture innovative outputs of R&D investment not accounted for by patents.

Combined, Tables 4 and 5 hint that trademark filing may be a precursor to innovative activity in the form of patenting or R&D, with trademark registration growing in importance amongst innovative firms. This lends some support to the theoretical argument that firms that engage in patent and R&D also invest in protecting the gains from innovation – potentially in the form of a higher quality product, better reputation, or larger customer base.

Figure 9 shows the distribution of firms based on the timing of patent and R&D activity

relative to the first trademark application filing. Each bar indicates the proportion of patent and trademark filers as a function of the date of their first patent filing relative to the date of the first trademark filing. The timing of patent activity relative to the timing of trademark activity is grouped into 5-year bins before and after time 0 – the reference point that indicates the firm filed for its first patent in the same year it filed for its first trademark registration. Nearly 10% of patent and trademark filers filed for their first patent and trademark registration in the same year. Approximately 25% of patent and trademark filers filed for their first patent 1-5 years after their first trademark filing, and around 17% filed for their first patent in the 1-5 years prior to their first trademark application. In fact, almost 50% of firms with both patent and trademark applications filed for their first patents and trademark registrations within a 5-year window, which strongly supports the notion that the two activities are intertwined. With respect to the overall timing of the two activities, in the majority of cases (59%) the first trademark application filing leads or coincides with the first patent filing. This result lends some support to the theoretical argument that firms that trademark also invest the returns from accumulated goodwill into product and process innovation.

5.4.2 Patenting and R&D spending before and after trademark filing

This section analyzes further the link between trademarks and innovation by examining issued patents and R&D expenditures by trademark filing firms before and after their first trademark filing compared a control group. It utilizes the same control group of non-trademark filing firms identified via propensity score matching as in the analysis of employment and revenue above.

Figure 10 plots the average number of patents before and after first trademark filing for firms that apply to register trademarks (treated) and the matching control firms (untreated). Mean patenting counts are again normalized to one for both groups at $t = -2$, two years before the first trademark filing. Trends in average patenting leading up to trademark filing are somewhat different. Average patenting increases for both groups in the year prior to filing. However, while the treated group exhibits a large increase in the year of filing, the control group's average patenting declines. After filing, average patenting grows at a similar rate for both treated and control firms, though the gap does expand after two years post-filing. Between $t = 2$ and $t = 5$, growth in the average number of patents is about 30% for the treated firms, compared to only about 15% for the control group.

The difference-in-difference model specified in (1) is used to analyze innovative activity for treated and control groups before and after the first trademark application. The last two columns of Table 3 report the estimation results for R&D expenditures and the number of issued patents as the dependent variables. For both the R&D expenditures and the number of patents, the treatment group has higher values for the pre-filing period, as indicated by the estimated value

of β_F. As in the case of employment and revenue, this finding suggests the likely presence of unobserved factors that result in higher patent and R&D activity for treated firms prior to trademark filing. In the post-filing period, average R&D expenditure and patenting by treated firms is higher than those by control firms, as the estimated values of β_{FP} suggest.

Overall, the results suggest that both selection and treatment effects associated with trademark filing are relevant for understanding firm-level innovative activity, as the theory suggests. Further investigation of both of these effects with a more stringent matching process to obtain a control group is left for future work.

6 Conclusion

The progress of empirical research on trademark activity by U.S. firms has been largely hampered by the lack of comprehensive firm-trademark linked data. This paper reports on the construction of a new firm-level longitudinal dataset that allows for the tracking of trademark filing activity over the life-cycle of a firm. The dataset brings together the USPTO's Trademark Case Files Dataset and the U.S. Census Bureau's Business Register and the Longitudinal Business Database for the period 1976-2014. Using the linked dataset, it is possible to identify if, and when, a firm first applies for a trademark. This key event is then tied to firm characteristics and dynamics to understand the nature of selection associated with trademark filing, as well as the treatment effects related to a trademark application.

The data is used to examine how firm employment, revenue, R&D expenditures, and patenting change after a firm's first trademark filing compared to their pre-filing levels. The analysis suggests the potential presence of strong selection and treatment effects associated with trademarks. Compared to the general population of firms, first-time trademark filers tend to be younger and larger firms. In addition, an initial analysis using a propensity score matching exercise indicates that average firm employment and revenue tend to be higher for firms after they file for their first trademark, compared to a control group. For first-time trademark filing firms, both patenting and R&D activity are also higher after the filing, again, compared to a control group.

The results also indicate that, while most of the firms that have applied for a trademark registration do not engage in innovative activity as measured by patent filings or grants, the proportion of trademark filing firms with a patent application is significantly higher than that of all firms. To the extent trademarks capture firm-level innovative activity not accounted for by patents, the relatively small presence of patents in trademark filing firms and the co-presence of trademark applications and R&D expenditure without any patenting is encouraging in terms of broadening the definition of the innovative segment of firms in the economy beyond simply those

that have patents. Nevertheless, a sizable fraction of firms with patents and R&D activity also tend to have trademark applications. This may indicate that trademark registrations are used by innovative firms to protect the potential gains from innovation. However, the reverse may also be true. Where trademark filing proceeds patent and R&D activity, firms may be investing the gains from accumulated goodwill into product and process innovation.

The trademark-firm linked dataset opens up several possibilities for future research. For instance, there is a large body of work in the marketing literature for which the dataset is highly relevant. The theoretical literature emphasize a connection between trademarks and customer acquisition and loyalty-building by firms. Various models focus on the role of trademarks in reducing consumer search and switching costs, establishing brand loyalty and goodwill, and signalling quality.[33] In general, by protecting a firm's investments in marketing and reputation-building, trademarks can lead to higher intensity of advertising and marketing expenditures, as trademark registrants can better appropriate the benefits from such expenditures. The new dataset can be instrumental in testing some of these theoretical implications.

There is also more to explore regarding the connection between trademarks and firm scale and scope. In particular, the role of trademark registration in new product introductions and changes in a firm's product portfolio can be examined. For instance, using changes in the narrowly-defined industry classifications for a firms' products before and after trademark filing, one can investigate whether trademark registrations facilitate scope expansion into products that are not closely related to a firm's core product portfolio.

Another avenue of research that can benefit from the new dataset is the valuation of trademarks. The dataset allows for the observation of first trademark registration by a firm, as well as its subsequent trademark registrations, and trademark reassignments. Reputation indicated by a trademark is a valuable asset that needs to be protected.[34] A reputable name or mark can also be traded.[35] The information contained in the trademark applications and assignments can be used, in conjunction with measures of firm value, to attach valuations to trademarks. In addition, the oppositions placed against a trademark filing and the resulting procedural outcomes observed in the dataset can be used to measure the inherent value of certain trademarks, as oppositions would be unlikely if the trademark was of little value.

[33] See, e.g., Landes and Posner (1987,1988) and Economides (1988) for theoretical arguments on the connection between trademarks and consumer behavior. For recent models of firm dynamics under costly and gradual customer acquisition, see Dinlersoz and Yorukoglu (2012) and Gourio and Rudanko (2014).

[34] See Cabral (2005) for a review of the theoretical literature on the economics of reputation. See also Cabral and Hortacsu (2010) and Cabral (2012) for studies of reputation on the Internet.

[35] See Tadelis (1999) for a theoretical analysis of reputation as a tradeable asset.

References

[1] Arora, Ashish, Bei, Xiaoshu, and Wesley M. Cohen. 2016. "Why Firms Trademark (or not): Evidence from the U.S. Trademark Data", Working paper, Duke University.

[2] Baroncelli, Eugenia, Carsten Fink, and Beata S. Javorcik. 2005. "The Global Distribution of Trademarks: Some Stylized Facts." *The World Economy*, 28: 765-782.

[3] Block, Jorn, Christian Fisch, Alexander Hahn, Philipp Sandner. 2015. "Why do SMEs File Trademarks? Insights from Firms in Innovative Industries", *Research Policy*, 44: 1915-1930. http://dx.doi.org/10.1016/j.respol.2015.06.007.

[4] Cabral, Luis. 2012. "Reputation on the Internet", in M. Peitz and J. Waldfogel (Eds), The Oxford Handbook of the Digital Economy, Oxford, UK: OUP, 343-354.

[5] Cabral, Luis, and Ali Hortacsu. 2010. "The Dynamics of Seller Reputation: Evidence from eBay", *Journal of Industrial Economics,* 58: 54-78.

[6] Cabral, Luis. 2005. "The Economics of Trust and Reputation: A Primer", Monograph in progress, New York University (available at http://pages.stern.nyu.edu/~lcabral/reputation/Reputation_June05.pdf)

[7] Decker, Ryan, John Haltiwanger, Ron S. Jarmin, and Javier Miranda. 2016. "Where Has All the Skewness Gone? The Decline in High-Growth (Young) Firms in the U.S.", *European Economic Review*, 86: 4-23.

[8] Davis, Steven, John Haltiwanger, Ron Jarmin, C.J. Krizan, Javier Miranda, Al Nucci and Kristen Sandusky. 2007. "Measuring the Dynamics of Young and Small Businesses: Integrating the Employer and Nonemployer Universes firms" NBER Working Paper No. 13226.

[9] Davis, Steven, Haltiwanger, John, and Scott Schuh. 1996. *Job Creation and Destruction.* Cambridge, MA: The MIT Press.

[10] Dinlersoz, Emin, and Mehmet Yorukoglu. 2012. "Information and Industry Dynamics", *American Economic Review*, 102: 884-913.

[11] Economides, Nicolas. 1988. "The Economics of Trademarks", *The Trademark Reporter*, 78: 523-539.

[12] Fazio, Catherine, Jorge Guzman, Fiona E. Murray, and Scott Stern. 2016. "A New View of the Skew: A Quantitative Assessment of the Quality of American Entrepreneurship", In *Kauffman Foundation New Entrepreneurial Growth*, Kansas City, MO: February 2016.

[13] Flikkema, Meindart, Ard-Pieter De Man, Carolina Castaldi. 2014. "Are Trademark Counts a Valid Indicator of Innovation? Results of an In-Depth Study of New Benelux Trademarks Filed by SMEs", *Industry and Innovation*, 21: 310-331.

[14] Foster, Lucia, Grim, Cheryl, and Nikolas Zolas. 2016. "A Portrait of Firms that Invest in R&D." U.S. Census Bureau Center for Economic Studies Paper No. CES-WP-16-41.

[15] Goldschlag, Nathan and Javier Miranda. 2016. "Business Dynamics Statistics of High Tech Industries", U.S. Census Bureau Center for Economic Studies Paper No. CES-WP- 16-55.

[16] Gourio, Francois and Leena Rudanko. 2014. "Customer Capital", *Review of Economic Studies*, 81, Vol. 3.

[17] Graham, Stuart, Galem Hancock, Alan Marco, and Amanda F. Myers. 2016. The USPTO Trademark Case Files Dataset: Descriptions, Lessons, and Insights.

[18] Graham, Stuart, Grim, Cheryl, Islam, Tariq, Marco, Alan, and Javier Miranda. 2015. "Business dynamics of innovating firms: linking U.S. patent data with administrative data on workers and firms" Working paper CES-WP-15-19, Center for Economic Studies, U.S. Census Bureau.

[19] Greenhalgh, C., M. Longland, and D. Bosworth. 2003. "Trends and Distribution of Intellectual Property: U.K. and European Patents and U.K. Trade and Service Marks, 1986-2000." *Report for the U.K. Intellectual Property Office.*

[20] Greenhalgh, C., and M. Rogers. 2008. "Intellectual Property Activity by Service Sector and Manufacturing Firms in the U.K., 1996-2000." *The Evolution of Business Knowledge,* Oxford University Press.

[21] Haltiwanger, John, and Ron Jarmin. 2007. "Integrated Longitudinal Business Database: Data Overview." 2007 Kauffman Symposium on Entrepreneurship and Innovation Data

[22] Haltiwanger, John, Jarmin, Ron, Kulick, Robert, and Javi Miranda. 2016. "High Growth Young Firms: Contributions to Job, Output, and Productivity Growth", forthcoming in *Measuring Entreprenurial Businesses: Current Knowledge and Challenges*, NBER Book Series Studies in Income and Wealth.

[23] Heath, Davidson and Christopher Mace. 2017. "What's a Brand Worth? Trademark Protection, Profits and Product Quality", Working Paper, University of Utah.

[24] Jensen, P. H. and E. Webster. 2004. "Patterns of Trademarking Activity in Australia." *Australian Intellectual Property Journal*, 15: 112-126.

[25] Jensen, P. H. and E. Webster. 2006. "Firm Size and the Use of Intellectual Property Rights." *Economic Record*, 82: 44-55.

[26] Landes, William, and Richard Posner. 2003. *The Economic Structure of Intellectual Property Law*. Cambridge, Mass.: Harvard University Press.

[27] Landes, William and Richard Posner. 1988. "The Economics of Trademark Law," *The Trademark Reporter*, 78: 267-306.

[28] Landes, William and Richard Posner. 1987. "Trademark Law: An Economic Perspective," *Journal of Law and Economics*, 30: 265-309.

[29] Sandner, Philipp. 2009. "The Identification of Trademark Filing Strategies: Creating, Hedging, Modernizing, and Extending Brands." *The Trademark Reporter*, 99: 1257-1298.

[30] Sandner, Phillipp and Joern Block. 2011. "The Market Value of R&D, Patents, and Trademarks." *Research Policy*, 40: 969-985.

[31] Tadelis, Steve. (1999), "What's in a Name? Reputation as a Tradeable Asset", *American Economic Review,* 89: 548–563.

[32] Zolas, Nikolas, Lybbert, Travis, and Prantik Battacharyya. 2016. "An 'Algorithmic Links with Probabilities' Concordance for Trademarks with an Application Towards Bilateral IP Flows", forthcoming in *The World Economy*, doi: 10.1111/twec.12382.

A Data construction

Because there are no disambiguated identifiers, such as an Employer Identification Number (EIN), that are shared between the TCFD and the BR, the two datasets have to be brought together using name and address matching techniques. The current matching effort is focused on matching domestic businesses observed in the trademark ownership database to the employer firm universe. Future work will incorporate businesses found in the trademark assignments database, which captures the transfer of trademarks between businesses, foreign trademark filing firms with domestic establishments, and non-employer trademark filing firms. The final output of matching methodology will be firm-level links between Census Bureau data and records in the TCFD.

The input frame for the matching process is the ownership databases in the TCFD. Several conditions are used to subset the raw owner file records that will be considered for matching to the BR. First, the filing, registration, and publication dates are used to exclude all trademarks for which the maximum year is less than 1976, the earliest year of the BR. Second, the country field is used to identify domestic trademark owners. Finally, the owner data file also contains certain types of businesses that are unlikely to be covered in the BR. These include entities representing federal and state governments as well as individuals. The owner entity type code is used to exclude these cases from the match to the BR. Table A1 shows the count of unique trademarks and owner records after applying each restriction to the sample.[36] The owner file initially contains more than 7 million trademarks and 17 million owner records. This reduces to almost 5 million trademarks and 12 million owner records after excluding foreign, federal, state, and individuals and those before 1976.

Table A1: Input Frame

Input Frame	Trademarks (1,000s)	Ownership Records (1,000s)
Owner File	7,214	17,381
Excluding Pre 1976	6,907	16,937
Excluding Foreign	5,939	14,407
Excluding Federal, State, Individual	5,048	12,289

Source: USPTO Trademarks Casefile Database, author's calculations.

Notes: Counts in thousands.

The first step in the matching process is the extraction of all unique name and address combinations from the owner data file. The owner data file contains an observation for each owner

[36] Each trademark has one or more "owner records", or records in the ownership database. Each record captures a different stage of the application, review, and registration process.

recorded for each trademark application, registration, and publication from 1870 to 2014.[37] Name and address information are collected at different times during a trademark's lifecycle often for the same business entity. Not only are there multiple instances of the name and address information for one or more businesses associated with a trademark, there are also different types of business names. For example, each record in the TCFD may include "former", "doing business as", and "composed of" business names. Each owner record also includes two street address variables, which correspond to the first and second lines of the owner's street address. In many cases the owner's full street address is split across these fields. It is not always the case, however, that line 1 should precede line 2. For example, line 1 might include the suite number while line 2 contains the street address or vice versa. In order to maximize the chances of identifying a match for each business in the TCFD, each unique name (across all name types) is combined with combinations of line 1 and line 2 of the street address (i.e. line 1, line 2, line 1 concatenated with line 2, and line 2 concatenated with line 1). This process produces one or more name and address combinations that are at-risk for a match to the BR for each owner record.

The next step in matching the TCFD data to the BR is the cleaning and preparation of both datasets. First, common strings that provide little identifying information for matching are removed from both datasets. These include symbols and punctuation (e.g. "& ", ".", "@"), common words (e.g. "and", "the"), and abbreviations or designations (e.g. "Co.", "LLC", "LTD"). Additional standardization procedures are used to standardize the name, street, and city fields.[38] Values in the state field are cleaned and standardized and the zip code field is subset to five digits. These cleaning algorithms are applied to the name and addresses from both the TCFD and the BR. Once cleaned, the name and addresses from both datasets are matched using fuzzy string techniques combined with a blocking methodology.

The matching of the TCFD to the BR proceeds in several steps. First, an initial subset of potential matches is identified based on a relatively loose name-only match criteria between all unique name and addresses extracted from the TCFD and all establishments in the BR.[39] All matches not meeting this very loose criteria are excluded. Matches of different quality from this initial match set are extracted based on combinations of name and address fields. These match categories are based on whether the match is on name in conjunction with different address fields (street, city, state, and zip code). Next, a string comparator is used to further clean and subset the matches. The Jaro-Winkler (JW) string comparator, which takes values ranging from zero and one as a function of how similar two strings are, is calculated for the TCFD and BR name and city pairs. Additionally, a composite JW score is calculated across all three fields. Name-

[37] As noted above, we exclude records filed, registered, and published prior to 1976.

[38] These standardization procedures include algorithms found in the the SAS Data Quality suite.

[39] This and other fuzzy name matches are done using SAS Data Quality algorithms. Where noted we use the Jaro-Winkler string comparator to clean matches.

only matches, which tend to be of the lowest quality, and matches made using only the name and a single address variable are kept only if the JW score between the name fields is greater than 0.85.[40] Across all match passes, only the highest quality pass is kept for each TCFD name and address. Among the remaining matches, the composite JW score is used to select only the highest quality matches.

The next step of the matching aggregates the establishment-level results from the BR match described above to the target firm-level match and integrates those matches with the LBD. From the LBD, the first and last year of observation for each firm ID are extracted. All unique combinations of TCFD name and address and firm ID are kept. The first and last firm years, in combination with the minimum and maximum trademark filing years associated with each TCFD name and address, are used to clean firm-level matches. Since trademark filing can plausibly occur well before a firm enters the employer universe, all matches that occur within a three year window before and after the firm's first and last year observed in the LBD are kept.

The final stage of the cleaning algorithm involves additional disambiguation of business names in the TCFD in order to increase the number of unique firm-TCFD matches. First, additional name standardization is performed to group matched and unmatched cases. Information from the LBD is then leveraged to further reduce multiple matches. In eliminating certain duplicate matches, matched firms are required to have positive employment, and cases where a firm has more than 10 trademarks *and at the same time*, the number of trademarks exceed the number of employees, are also dropped. Table A2 shows the match quality distribution for the resulting linked data. The majority (81%) of matches use a combination name, street, city, and state. Note also that less than 10% of the matches rely on name only, which is a relatively weak criteria that will tend to generate more false positives compared to matches with address blocking.

In order to better understand why TCFD name and addresses do not receive a firm match, unmatched cases are matched to the non-employer business register (ILBD). The ILBD is based on administrative data on income tax returns (Form 1040 with attached Schedule C(s)). In the ILBD some of the observations include individual's names, while others pertain to firm names. To carry out the match, an algorithm is applied to determine whether an observation likely represents an individual's name. Steps generally similar to the match to the BR are then followed to match the unmatched trademarks to the name and addresses appearing in the ILBD. For this study, these matches are used primarily to better understand the quality of matches to the employer universe. Future work will investigate matches to non-employer businesses and their growth and transition dynamics.

Table A3 shows the match rates between trademarks in the TCFD owner file and the LBD

[40] This and other cutoff values were reached by visual inspection of the matches and JW scores. The score of 0.90 balanced Type I and Type II errors.

Table A2: Match Quality Distribution

Match Pass	Share of Matches
Name, Street, City, State	81.3
Name, State, Zip Code	4.8
Name, Zip Code	0.0
Name, State	5.1
Name Only	8.7

Source: USPTO Trademarks Casefile Database, Business Register, author's calculations.

Notes: Match passes listed in order of decreasing quality.

and ILBD by decade. This table also reports the percent of employer matches that are unique. The overall match rate to the LBD is over 83% and 2.4% to the ILBD. The first point to note is that the number of trademarks filed each year has grown substantially over time. The match rate to the employer universe has fallen about 16 points between the 1980s and the 2010s. The percent of matches that are unique is increasing over time, rising from 59% in the 1980s to 88% in the 2010s. The match rate to the nonemployer universe, in contrast to the employer match, has been growing since 1990.

Table A3: Match Rates by Decade

		LBD		ILBD
Years	Trademarks	Match Rate	Percent Unique	Match Rate
1980-1989	363,000	88.8	59.0	
1990-1999	582,000	88.8	64.6	1.4
2000-2009	1,058,000	83.9	75.2	2.7
2010-2014	649,000	73.3	87.7	3.8

Source: Observation counts rounded. USPTO Trademarks Casefile Database, LBD, author's calculations.

Notes: Decades are defined by trademark registration year. Percent unique is the share of matches that are to a single firmid. The range of the BR and LBD is 1976 to 2014 and the range of the ILBD is 1993-2014.

Analysis of the underlying matched and unmatched records reveals several patterns.[41] First, the decline in match rates begins in the late 1990s and speeds up around 2010. As shown in Figure

[41] For this analysis, only the registered trademarks are considered, rather than applications for trademarks, since registered trademarks require demonstration of commercial use and thus may be better associated with

A1, the decline is most pronounced among records identified as "Limited Liability Company" (LLCs) in the TCFD. The match rate for LLCs falls from about 91% in the early 1990s to less than 45% in 2014. In addition, LLCs are the fastest growing entity type in the TCFD over this period rising from about 2k filings in 1990 to almost 100k filings in 2014. The fraction of LLCs has also been growing over this period in the general population of firms.

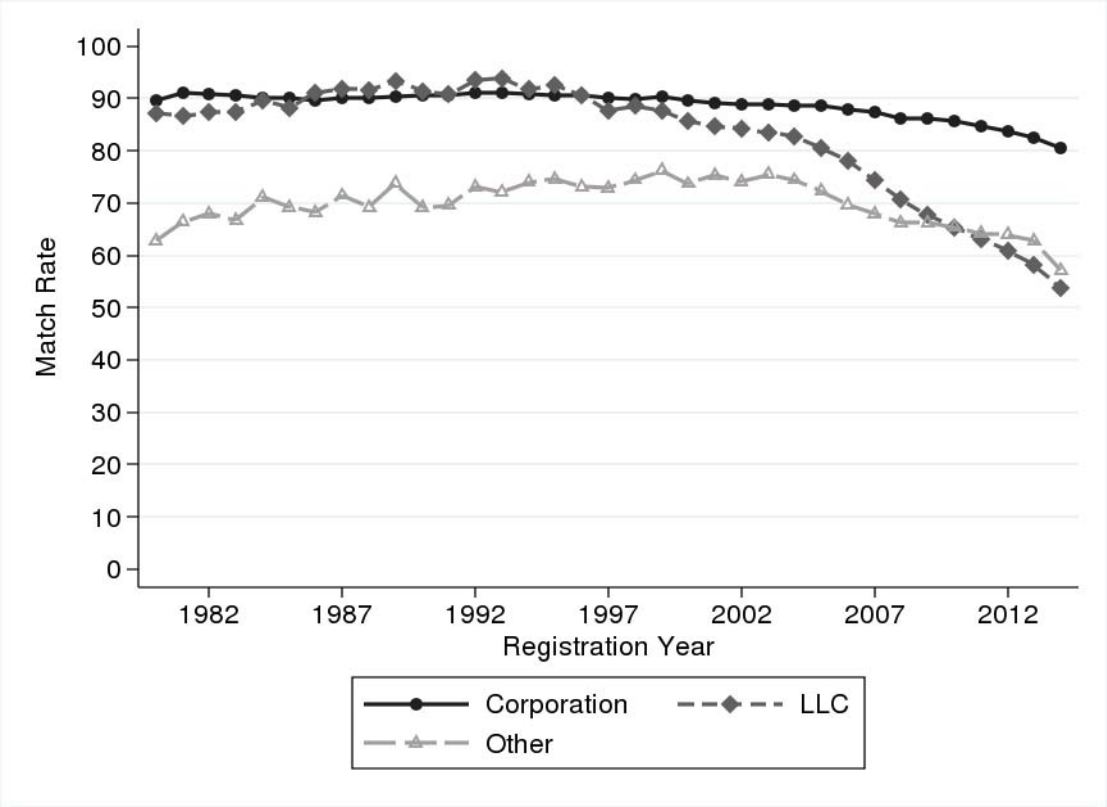


Figure A1: LBD Match Rate by Legal Form of Organization

Source: USPTO Trademarks Casefile Database, LBD, author's calculations.

Notes: Only registered trademarks included. Corporations and LLCs are identified using the legal entity type code found in the TCFD.

To investigate the declining match rate between the TCFD and the LBD, several hypotheses for the decline are explored, including deteriorating string quality, a compositional shift towards nonemployers, and the growth of informal businesses. First, if the quality of the string variables in the TCFD (e.g. name and address fields) declines over time this could adversely affect the match rate to the LBD. To explore this hypothesis, measures of mean string length are constructed over time for both names and addresses. The average name length and match rate exhibit very similar time series patterns with a simple correlation of 0.96. Moreover, the average string length

employer businesses in BR and LBD.

of the name variable falls from almost 24.5 characters to less than 22.5, which could represent an 8% loss of information over the period. Figure A2 shows the relationship between string length and match rate over time. Both relatively short and long string names have lower match rates. Importantly, the match rate declines over time for all string lengths. This implies that the decreasing information in the TCFD name field is unlikely to account for the declining match rate to the LBD.

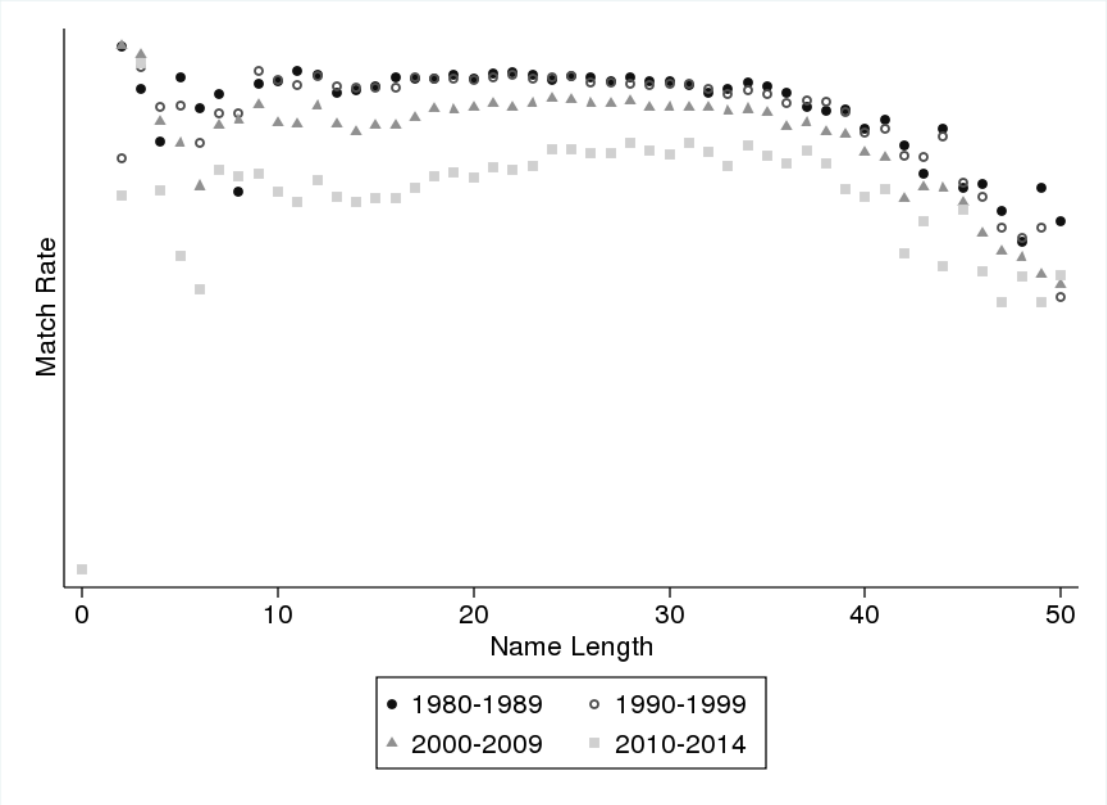


Figure A2: Match Rate and String Length

Source: USPTO Trademarks Casefile Database, LBD, author's calculations.

Notes: Match rates on the y-axis suppressed to avoid the disclosure of sensitive information.

Another potential explanation for the decline in match rates would be a compositional shift towards nonemployers among businesses filing for trademarks. As shown in Table A3 the match rate to the ILBD does increase over time. However, the increased matches to the nonemployer universe is not enough to make up for the decline in the LBD match rate. Assuming all TCFD businesses not found in the LBD were true non-employers, the ILBD match rate would need to be at least five times larger to fill the gap in the falling employer match rate.[42] It is also the case

[42] As an additional validation we compare the share of nonemployers specifically among LLCs in the BR from 2007 to 2014, years for which we are able to distinguish LLCs. The share of nonemployers among LLCs is actually

that the vast majority of records in the ILBD are person records derived from tax filings for sole proprietorships. These records often have the individual filer's name as the name of record in the ILBD. Manual inspection of the unmatched cases later in the time series suggests that the vast majority of unmatched cases include business names rather than person names.

If a growing share of trademark application filings are associated with informal or not yet implemented business ideas, one would expect the match rate to employer firms to fall over time. A number of the unmatched cases filed in recent years appear to have a web presence on platforms such as Etsy or Facebook. It could be the case that these types of businesses either do not earn enough revenue to file a Schedule C or the businesses were operated only sporadically. According to the 2017 IRS instructions for form 1040, self-employed individuals that had net earnings of less than $400 were not required to file a Schedule C and would therefore not appear in the ILBD. Additionally, the 2017 IRS instructions for form Schedule C clarify that a "sporadic activity or a hobby does not qualify as a business". With trademark application fees as low as $275 for electronic applications, it is possible that some individuals file for trademarks in order to protect the potential exploitation of a business idea that otherwise remains a hobby.[43] As with a compositional shift towards nonemployer businesses, a growing share of informal businesses in the TCFD would prove the declining match rate to the BR to be innocuous.

Finally, the quality of the matches are analyzed. A random sample of over 350 matches are classified as either true or false positives by clerical review. Once classified, the precision of the matching methodology can be measured. The precision measure, commonly used to evaluate the quality of information retrieval algorithms, captures the proportion of matches that are true matches. The match turns out to be fairly precise, with an overall precision score of 94%, meaning that 94% of the matches represent true matches based on the sample analyzed. The precision of matches is relatively stable over time falling slightly by the end of the period.

Name and address matching techniques result in robust linkages between the information in TCFD and Census Bureau information on businesses in the BR and LBD. The TCFD is relatively large, containing over 5 million trademarks. Over the period, about 83% of these trademarks are matched to at least one firm ID. While the match rate somewhat declines over time, the ambiguity of those matches, that is how many TCFD records end up being linked to multiple firm ID's, also declines. The decline in match rate is concentrated among records flagged as LLCs in the TCFD. Several potential explanations for the decline in the employer match rate are explored, but none are able to entirely explain the patterns observed in the data. Ultimately,

falling in the BR over time, a finding inconsistent with a compositional shift towards nonemployers among the universe of LLCs driving the decline in match rates.

[43] Trademark applications may be filed as "use in commerce" or "intent to use in commerce". In either case, in order for an application to obtain registration the USPTO requires proof that a trademark is used in commerce. See Graham, Hancock, Marco, and Myers (2013) for details.

matches prove to be of high quality, with a precision score over 90%.

Table 1. Firm size (employment) and age for trademark filing firms versus others

	Firms with a TM filing		Firms with a current year TM filing		Firms with first TM filing		Firms with no TM filing		Average no. of
Year	Mean Size	Mean Age	Mean Size	Mean Age	Mean Size	Mean Age	Mean Size	Mean Age	TMs per firm
1997	100.1	7.8	190.5	6.2	75.9	4.0	9.5	6.5	2.3
1999	102.0	8.3	180.2	6.2	50.2	4.0	9.7	7.0	2.4
2001	101.3	8.9	214.3	7.2	54.9	4.6	10.0	7.6	2.3
2003	93.5	9.7	200.5	7.8	50.9	4.8	9.6	7.9	2.2
2005	89.6	10.3	196.7	8.1	47.8	4.9	9.5	8.2	2.3
2007	89.8	10.7	186.5	8.4	52.7	5.2	9.7	8.7	2.2
2009	84.5	11.5	188.9	9.3	48.0	5.9	9.5	9.6	2.1
2011	81.5	12.2	186.5	9.6	46.6	5.7	9.4	10.2	2.2
2013	83.9	13.8	200.8	11.6	57.5	8.3	9.7	10.8	2.2

Table 2. Firm size (employment) and age for trademark filing firms versus others - sectoral differences

	Firms with a TM filing		Firms with a current year TM filing		Firms with first TM filing		Firms with no TM filing	
Industry	**Mean Size**	**Mean Age**	**Mean Size**	**Mean Age**	**Mean Size**	**Mean Age**	**Mean Size**	**Mean Age**
Agriculture, Mining and Utilities	203.5	14.4	536.7	11.0	181.3	8.0	11.3	11.3
Construction, Transportation and Warehousing	92.8	14.4	238.7	12.2	81.6	10.2	7.4	10.7
Manufacturing	113.9	16.7	282.2	14.4	53.3	8.8	17.5	13.4
Wholesale	43.6	15.0	99.4	12.8	38.1	8.5	8.5	12.1
Retail	68.3	12.5	201.1	9.9	19.4	7.4	7.9	10.5
Information	76.6	13.0	256.6	10.9	61.5	6.5	8.6	9.4
Finance and Real Estate	88.5	14.0	233.8	12.3	64.4	9.5	5.2	10.7
Professional Services	39.3	12.1	84.5	9.3	43.5	7.0	5.5	10.3
Management and Admin. Support	153.8	14.1	345.9	12.0	116.9	9.0	15.8	9.9
Education and Healthcare	210.4	14.3	487.9	12.2	126.8	9.4	15.7	11.6
Entertainment, Accommodation and Food	94.0	12.2	212.3	9.9	44.8	7.6	15.5	8.5
Other Services	32.6	14.7	68.1	12.8	25.3	9.5	5.7	13.0

Table 3. Regression analysis of the relation between various firm outcomes and the first trademark filing

	Dependent Variable			
Independent Variables	**Employment**	**Revenue**	**R&D Exp.**	**Patents**
$FILE_i$	0.143^{***}	0.323^{***}	0.406^{***}	0.011^{***}
	[0.015]	[0.023]	[0.121]	[0.002]
$POST_FILE_{it}$	0.997^{***}	0.197^{***}	0.191^{*}	0.001
	[0.018]	[0.011]	[0.075]	[0.001]
$FILE_i \times POST_FILE_{it}$	0.297^{***}	0.195^{***}	0.783^{***}	0.013^{***}
	[0.008]	[0.013]	[0.133]	[0.001]
$CONSTANT$	6.459^{***}	7.376^{***}	5.108^{***}	-0.011^{**}
	[0.032]	[0.018]	[0.069]	[0.003]
R^2	0.24	0.18	0.41	0.05
N	$9.5M$	$5M$	$80K$	$10M$

Notes: All regressions include industry (4-digit NAICS) and year fixed effects. All dependent variables are transformed using hyperbolic sine transformation. Standard errors clustered by industry are in parentheses. *,**,*** indicate significance at 10, 5, and 1% respectively. N is rounded to avoid disclosure. Each control firm is weighted by the total number of control firms for the corresponding treated firm.

Table 4. Patent application and R&D activity
for all firms versus firms with trademark filing

All U.S. Firms

Year	Firms	% of Firms with Patents	% of Firms with R&D	% of Firms with Patents & R&D
1997	4,700,000	0.40	0.17	0.10
1999	4,900,000	0.46	0.18	0.11
2001	4,900,000	0.51	0.19	0.12
2003	5,200,000	0.53	0.23	0.14
2005	5,500,000	0.53	0.29	0.16
2007	5,600,000	0.54	0.32	0.17
2009	5,300,000	0.58	0.42	0.22
2011	5,300,000	0.60	0.47	0.24

Firms with Trademark Filing

Year	Firms	% of Firms with Patents	% of Firms with R&D	% of Firms with Patents & R&D
1997	126,000	8.33	3.33	1.98
1999	154,000	8.44	3.31	1.95
2001	169,000	9.17	3.43	2.07
2003	182,000	9.34	4.07	2.42
2005	202,000	9.41	4.85	2.87
2007	227,000	9.03	5.07	2.91
2009	241,000	9.13	5.81	3.32
2011	254,000	9.06	6.69	3.66

Notes: The figures provide fractions of firms that have ever applied for a patent or performed R&D. Patent data comes from USPTO and R&D data from the BRDIS. Figures are rounded for disclosure purposes.

Table 5. Co-presence of trademark filing, patent applications, and R&D activity

Firms with Patent Applications

Year	Firms	% of Firms with Trademarks	% of Firms with R&D	% of Firms with Trademarks & R&D
1997	19,000	55.26	25.79	13.16
1999	22,500	57.78	24.44	13.33
2001	25,000	62.00	23.20	14.00
2003	27,500	61.82	25.82	16.00
2005	29,000	65.52	30.34	20.00
2007	30,500	67.21	31.80	21.64
2009	31,000	70.97	37.10	25.81
2011	32,000	71.88	39.06	29.06

R&D Performing Firms

Year	Firms	% of Firms with Trademarks	% of Firms with Patents	% of Firms with Trademarks & Patents
1997	8,000	52.50	61.25	31.25
1999	8,800	57.95	62.50	34.09
2001	9,200	63.04	63.04	38.04
2003	12,000	61.67	59.17	36.67
2005	16,000	61.25	55.00	36.25
2007	18,000	63.89	53.89	36.67
2009	22,000	63.64	52.27	36.36
2011	25,000	68.00	50.00	37.20

Notes: The figures provide fractions of firms that have ever applied for a patent or trademark, or performed R&D. Patent data comes from USPTO and R&D data from the BRDIS. Figures are rounded for disclosure purposes.

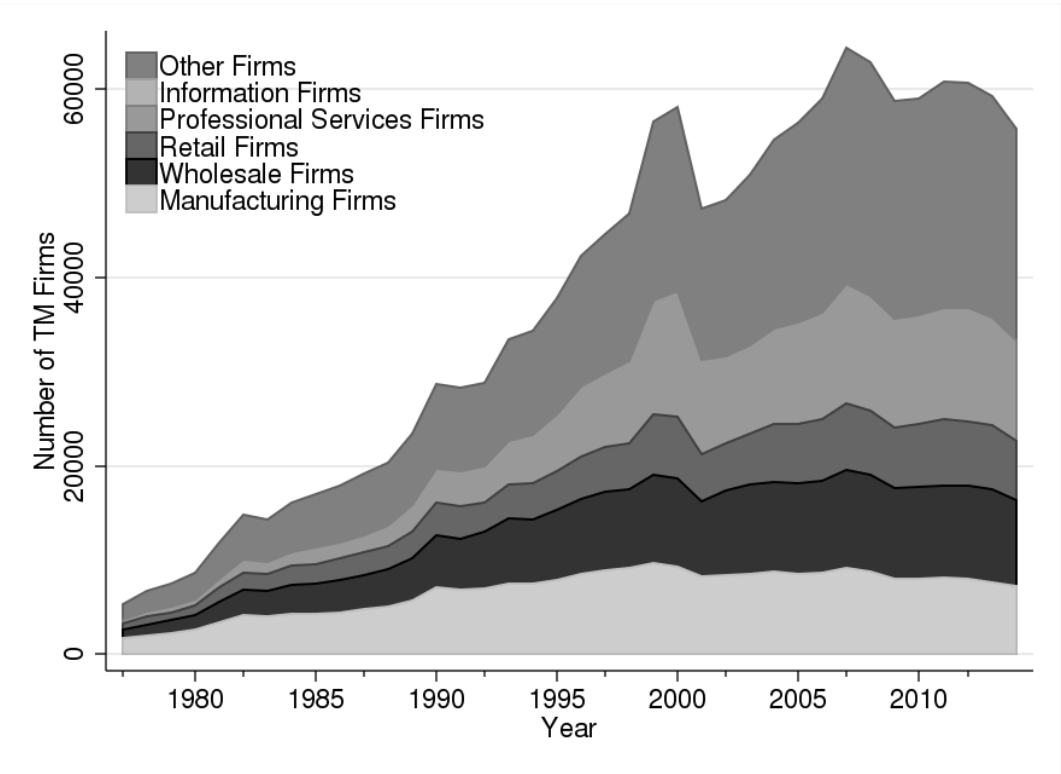


Figure 1: The number of firms that file for a trademark for the first time, by sector over time

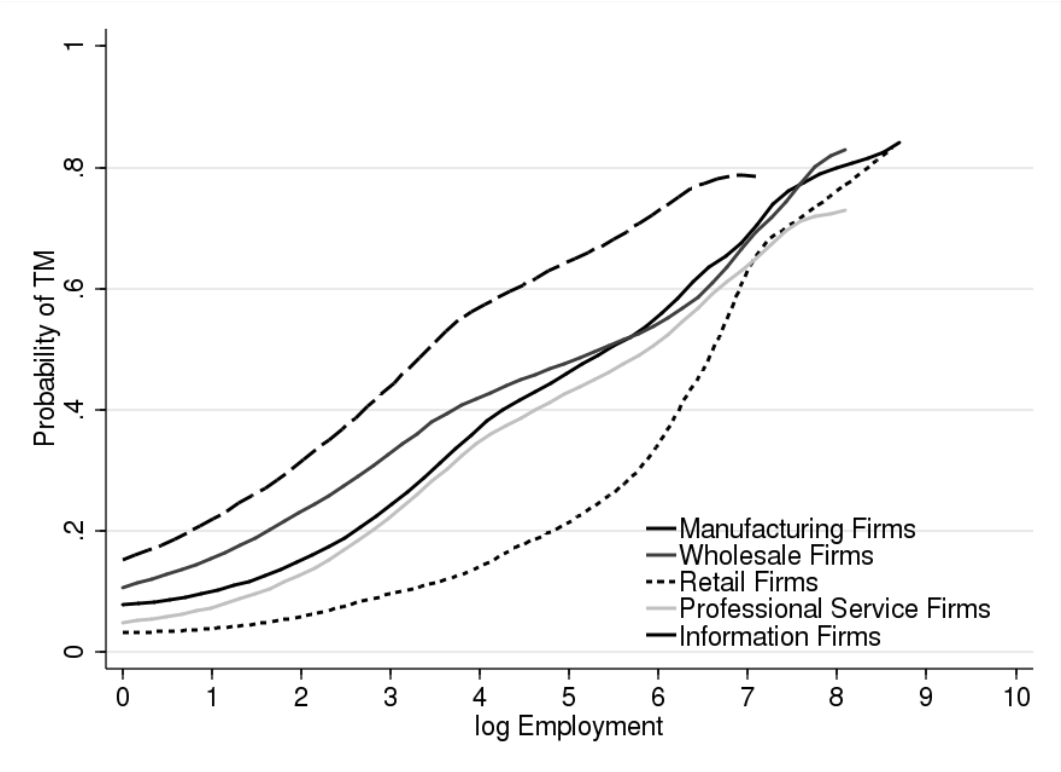


Figure 2: The probability that a firm has a trademark filing, by firm size and sector

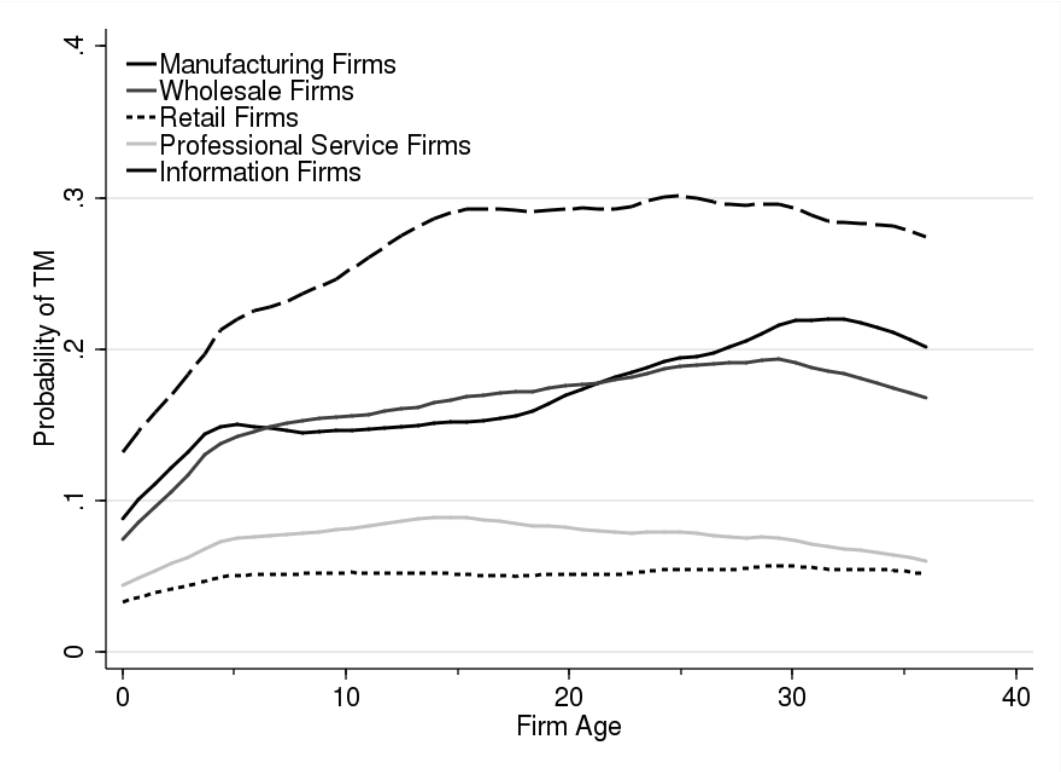


Figure 3: The probability that a firm has a trademark filing, by firm age and sector

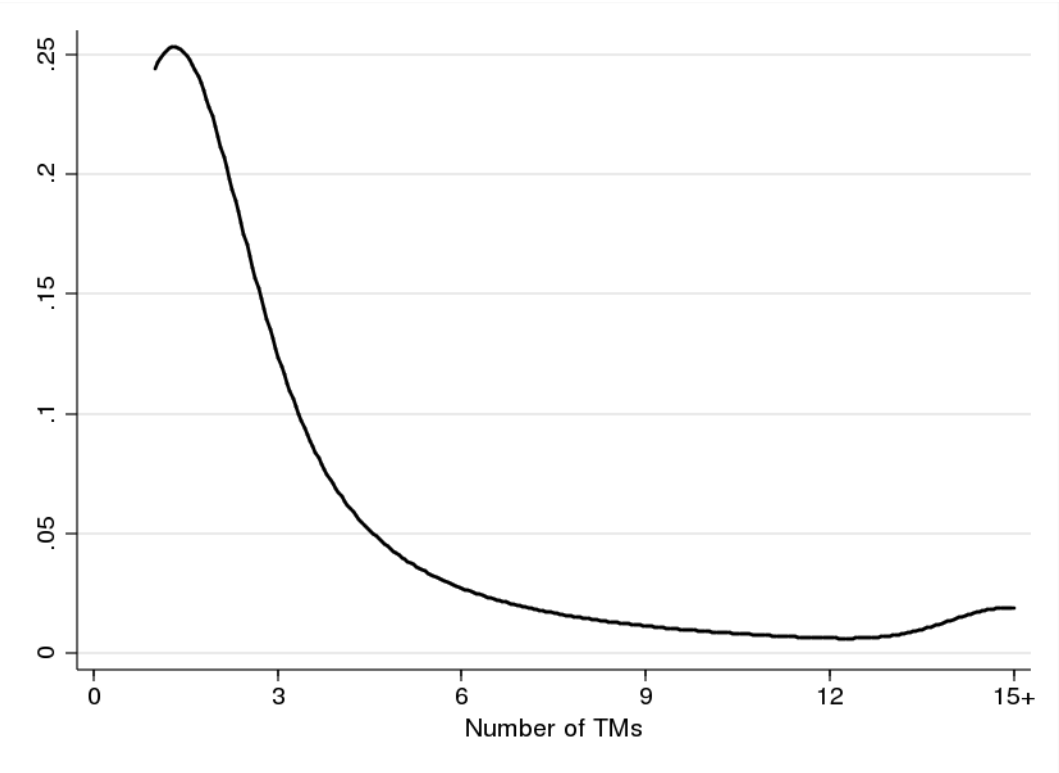


Figure 4: The distribution of firm trademark intensity (the number of trademark filings by a firm)

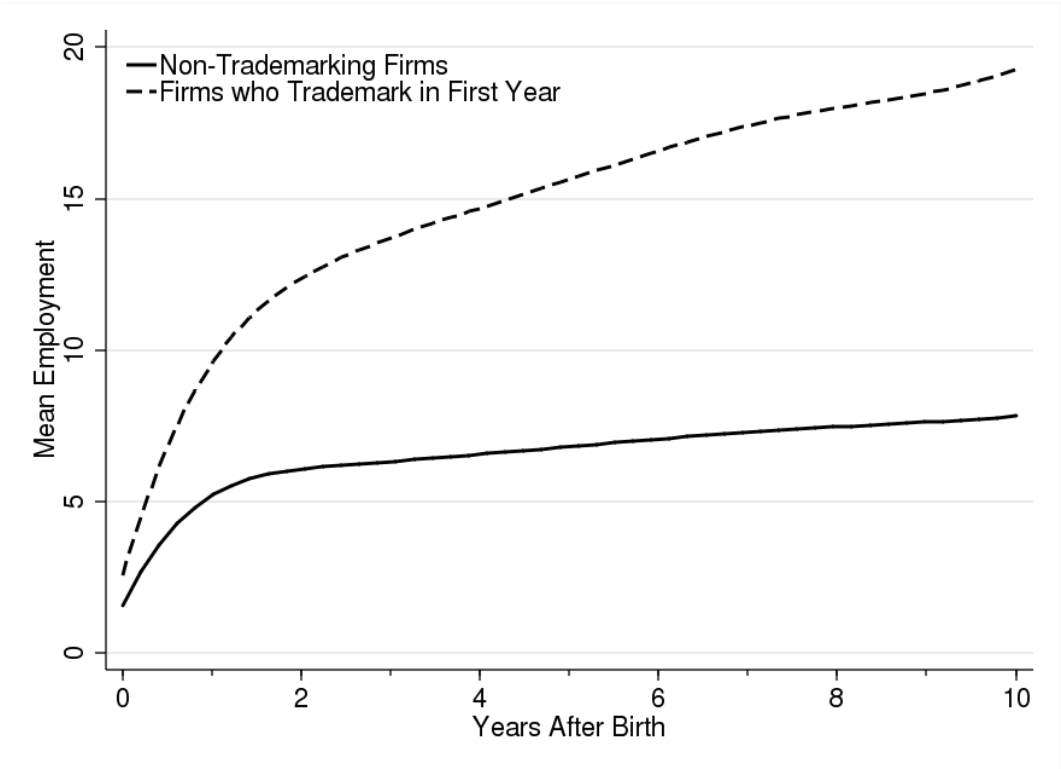


Figure 5: Average employment of new firms over time, by trademark filing status in the year of birth

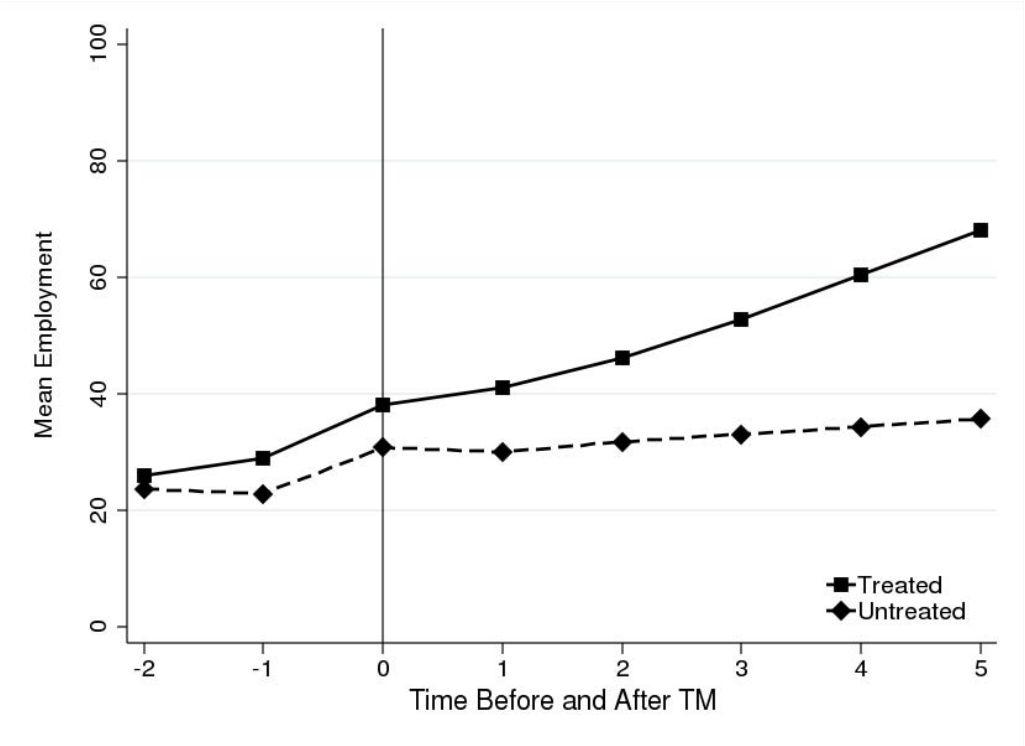


Figure 6: Average employment before and after the first trademark filing, the treated versus the control group

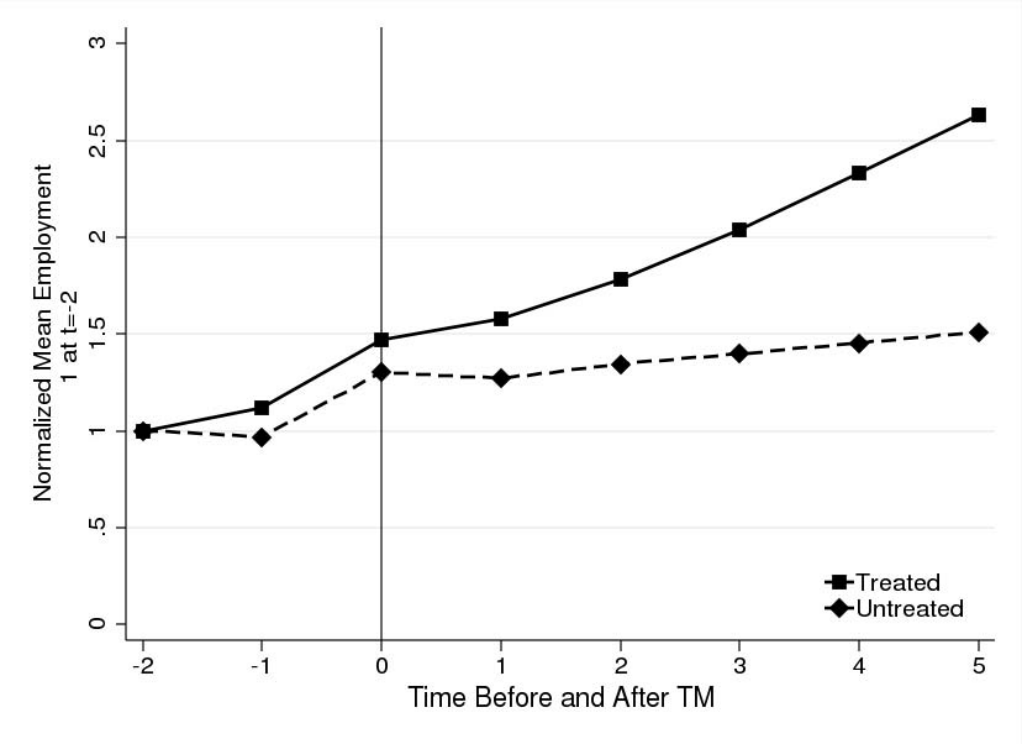


Figure 7: Normalized average employment before and after the first trademark filing, the treated versus the control group

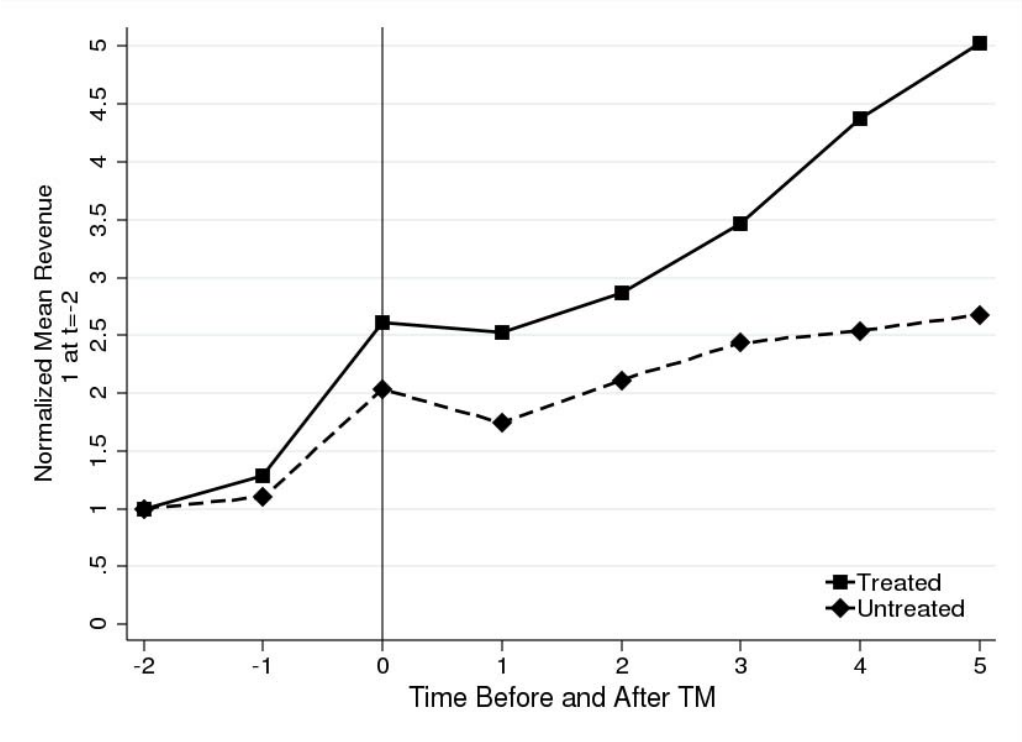


Figure 8: Normalized average revenue before and after the first trademark filing, the treated versus the control group

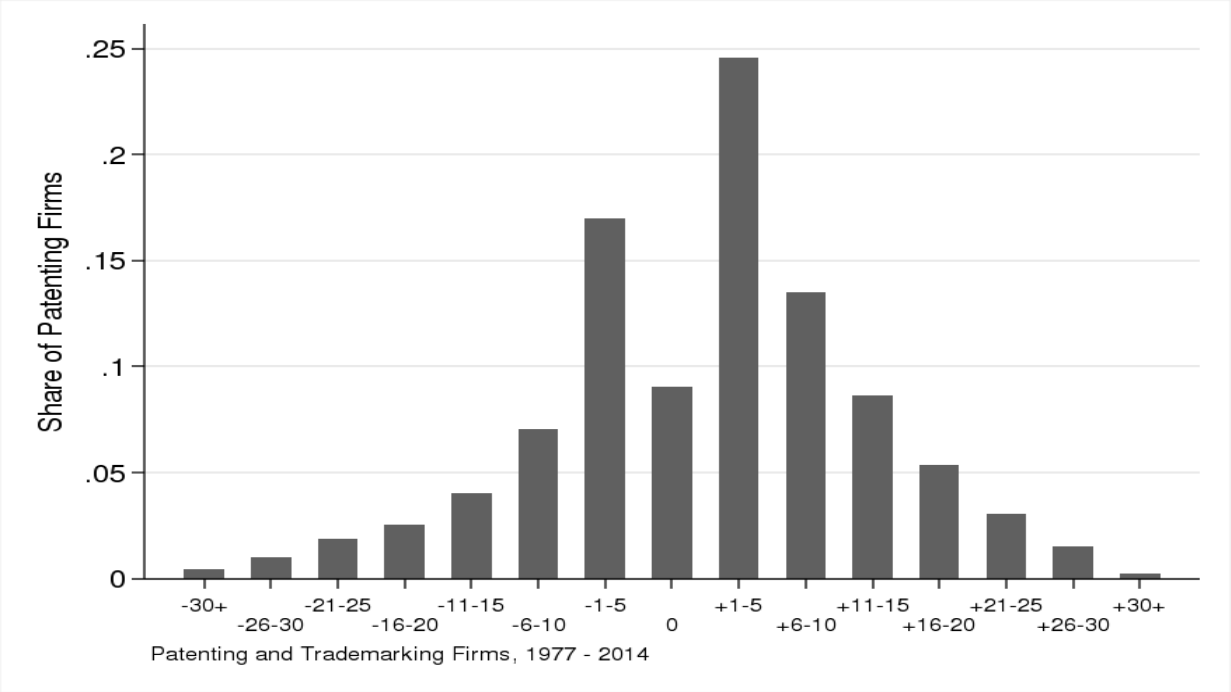


Figure 9: Timing of first patenting application relative to first trademark filing for firms which do both

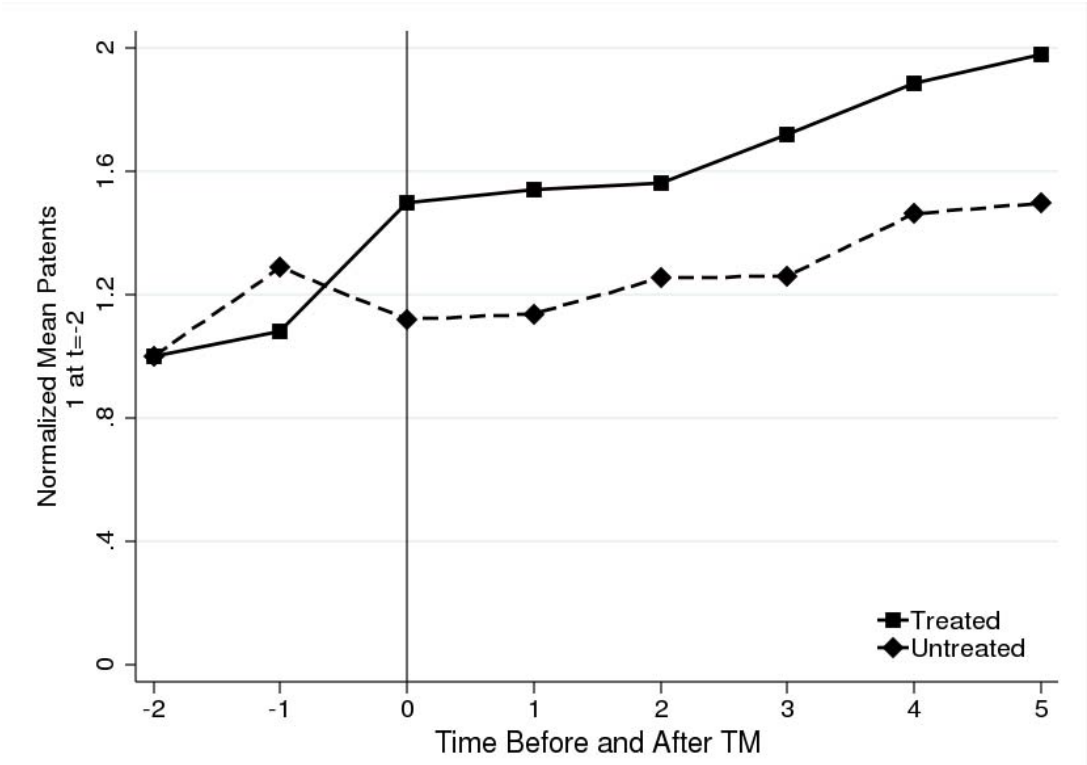


Figure 10: Normalized average number of patents before and after the first trademark filing, the treated versus the control group